

Special Holonomy and Ricci-Flat Metrics

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Abstract

In this thesis we focus on the following question: when a family of manifolds with special metrics degenerates, what does the metric look like near the singularities? A natural setting in which to study this question is that of manifolds with so-called special holonomy. We provide the background in Riemannian and complex geometry needed to discuss Yau's theorem, which provides the existence theory for Calabi-Yau metrics. In real dimension four the Calabi-Yau condition is equivalent to the existence of a closed triple of two-forms, and we give a complete proof of this equivalence, following an outline of Donaldson. The Gibbons-Hawking construction generates such triples from positive harmonic functions on open subsets of $\mathbb{R}^3$; importantly for this thesis it produces the Eguchi-Hanson metric which can be understood to model the simplest kind of singularity known as an ordinary double point. We compute the curvature of an arbitrary Gibbons-Hawking metric in closed form as outlined in [Don12, §5] and for the Eguchi-Hanson metric determine the behaviour at infinity. We then show that the curvature energy density of a Gibbons-Hawking metric is the divergence of an explicit vector field on $\mathbb{R}^3$, and compute the energy of the A_k spaces as a flux. Finally, we outline the use of the Eguchi-Hanson metric in gluing constructions: the Kummer construction, and Joyce's construction of the first compact G_2 -manifolds.

Declaration

The work contained in this thesis is my own work unless otherwise stated.

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CHAPTER 1

Introduction

1.1 A degenerating family of surfaces

Consider the family of surfaces $X_t \subset \mathbb{CP}^3$ given by the quartic equation

$$z_0^4 + z_1^4 + z_2^4 + z_3^2 q(z_0, z_1, z_2) = t z_3^4, \quad (1.1)$$

for a non-degenerate quadratic form q and a parameter $t \in \mathbb{C}$. For generic t (in particular for all sufficiently small $t \neq 0$) the surface X_t is a smooth quartic, hence a K3 surface, and by Yau's resolution of the Calabi conjecture carries a Ricci-flat Kähler metric. At $t = 0$ the surface acquires an *ordinary double point* at $[0 : 0 : 0 : 1]$: a point locally modelled on the quadric cone $\{z_1^2 + z_2^2 + z_3^2 = 0\} \subset \mathbb{C}^3$, isomorphic to the quotient $\mathbb{C}^2 / \{\pm 1\}$. *What does the Ricci-flat metric on X_t look like, near this singularity, as the family degenerates, i.e. as $t \rightarrow 0$?* To spare the reader suspense, this ordinary double point singularity can be modelled by the Eguchi-Hanson metric, which we examine from Chapter 6 onwards.

1.2 Special holonomy and explicit metrics

What is special holonomy, and why should we care? The Levi-Civita connection of a Riemannian manifold defines a canonical notion of parallel transport, and the holonomy group refers to the group of transformations of a tangent space obtained by parallel transport around loops. A famous result by Berger tells us that for a simply connected, irreducible, non-symmetric Riemannian manifold (M, g) the possible holonomy groups are $\mathrm{SO}(n)$, $\mathrm{U}(m)$, $\mathrm{SU}(m)$, $\mathrm{Sp}(m)$, $\mathrm{Sp}(m) \cdot \mathrm{Sp}(1)$, G_2 and $\mathrm{Spin}(7)$ (Theorem 2.10). In general, the study of such manifolds is of interest because all but the generic $\mathrm{SO}(n)$ case and the $\mathrm{U}(m)$ case force the metric to be *Einstein*, meaning that $\mathrm{Ric} = \lambda g$ for a constant λ , and indeed Ricci-flat except in the $\mathrm{Sp}(m) \cdot \mathrm{Sp}(1)$ case. In Chapters 3 and 4 we cover the existence theory for the $\mathrm{SU}(m)$ case, providing the complex geometry necessary to discuss the reframing of the Calabi conjecture as a complex Monge-Ampère equation and its resolution by Yau. In the second half of the thesis we study the simplest non-trivial examples

of these Calabi-Yau metrics. A useful reframing of the Calabi-Yau/hyperkähler condition, since $SU(2) \cong Sp(1)$, in real dimension four is that of a triple of closed two-forms satisfying a simple algebraic condition. In the literature this is known as a positive definite triple. The Gibbons-Hawking construction gives a way of producing such triples from positive harmonic functions on open subsets of $\mathbb{R}^3$ (Chapter 6): its first non-trivial output is the Eguchi-Hanson metric. In Chapter 7 we compute the curvature of the Gibbons-Hawking spaces in full, following the outline given in [Don12]: first for an arbitrary potential, then for Eguchi-Hanson. We then show that the curvature energy density of any Gibbons-Hawking metric is the divergence of an explicit vector field on $\mathbb{R}^3$, so that the total energy $\int |\text{Rm}|^2$ can be computed as a flux. Finally, in Chapter 8 we return to the degenerating family of K3 surfaces, and discuss the use of the Eguchi-Hanson metric in gluing constructions: the Kummer construction of K3 metrics and Joyce's construction of the first compact G_2 -manifolds.

1.3 Main results and attribution

Stated informally, the main computational results are as follows. For the Gibbons-Hawking metric of a positive harmonic function f on an open subset of $\mathbb{R}^3$ the full Riemann curvature tensor is the trace-free part of the matrix $(f_{ij}/f^2 - 3f_i f_j / f^3)$ acting on the anti-self-dual two-forms (Theorem 7.2). In the Eguchi-Hanson case, the curvature has eigenvalues in the ratio $(2, -1, -1)$ with $|\text{Rm}|$ constant on the confocal ellipsoids with foci at the two poles and decaying at rate $O(\rho^{-6})$ (Proposition 7.4). Finally, the curvature energy density of a Gibbons-Hawking metric is the divergence of an explicit vector field on $\mathbb{R}^3$ in terms of f and its first two derivatives (Theorem 7.8). The energy of the A_k spaces is evaluated as a flux in terms of this vector field (Theorem 7.9).

Proposition 5.1 is stated in [Don06, Don12], and the proof in Chapter 5 is the author's, following those outlines. The curvature formula of Chapter 7 follows [Don12, §5]. The author's supervisor suggested computing these curvatures explicitly and examining their behaviour at infinity. The resulting calculations, in particular the Eguchi-Hanson case (§7.2), are the author's own, and the flux computation of §7.3 was carried out independently of [dB17], where the curvature energy of a related Gibbons-Hawking metric is evaluated by a similar argument. The value of the energy for any ALE space follows from the Gauss-Bonnet theorem with boundary corrections [Nak90]. The divergence identity of Theorem 7.8 is equivalent to a formula for the curvature density of a Gibbons-Hawking metric going back to [NS96, eq. (32)] and stated in [GW00, Rmk. 2.4]. The derivation given here was obtained independently (see Remark 7.10).

1.4 Notation and conventions

The curvature tensor is $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]}Z$, with index convention $(R(e_a, e_b)e_c)^d = R^d_{cab}$, and the Ricci tensor is $\text{Ric}(X, Y) = \text{tr}(Z \mapsto R(Z, X)Y)$. A Hermitian metric and its fundamental form are related by $\omega(X, Y) = g(JX, Y)$, and a two-form σ and its associated skew endomorphism by $\sigma(X, Y) = g(A_\sigma X, Y)$. On an oriented Riemannian four-manifold, $\Lambda^\pm$ denote the bundles of self-dual and anti-self-dual two-forms, and $\sigma_1 = e^{01} + e^{23}$, $\sigma_2 = e^{02} + e^{31}$, $\sigma_3 = e^{03} + e^{12}$ the standard self-dual frame attached to an oriented orthonormal coframe (e^0, e^1, e^2, e^3) , with $e^{ab} := e^a \wedge e^b$.

CHAPTER 2

Special Holonomy

2.1 Quaternions and the group $\mathrm{Sp}(1)$

We follow [Sti08]. The quaternions $\mathbb{H}$ form the four-dimensional real algebra spanned by $1, i, j, k$ subject to the relations

$$i^2 = j^2 = k^2 = -1, \quad ij = k, \quad jk = i, \quad ki = j, \quad (2.1)$$

with $ji = -k$ and cyclic analogues, so that multiplication is associative but not commutative. A concrete matrix model represents $q = a + bi + cj + dk$ as

$$q = \begin{pmatrix} a + id & -b - ic \\ b - ic & a - id \end{pmatrix}, \quad (2.2)$$

under which quaternion multiplication becomes matrix multiplication and the squared norm $|q|^2 = a^2 + b^2 + c^2 + d^2$ is the determinant, so $|pq| = |p||q|$.

DEFINITION 2.1. $\mathrm{Sp}(1)$ is the group of unit quaternions, $\{q \in \mathbb{H} : |q| = 1\}$, which is diffeomorphic to S^3 . In the matrix model above, the unit quaternions are the special unitary 2×2 matrices, giving the isomorphism $\mathrm{Sp}(1) \cong \mathrm{SU}(2)$.

By identifying $\mathbb{R}^3$ with the imaginary quaternions $\mathrm{Im} \mathbb{H} = \mathbb{R}i + \mathbb{R}j + \mathbb{R}k$ one sees that conjugation by a unit quaternion preserves this subspace.

LEMMA 2.2. *For $t \in \mathrm{Sp}(1)$, the map $q \mapsto tq\bar{t}$ is a rotation of $\mathrm{Im} \mathbb{H}$. Writing $t = \cos \theta + u \sin \theta$ with $u \in \mathrm{Im} \mathbb{H}$ a unit vector, this rotation is through angle 2θ about the axis u . The resulting homomorphism*

$$\mathrm{Sp}(1) \longrightarrow \mathrm{SO}(3) \quad (2.3)$$

is surjective with kernel $\{\pm 1\}$, so rotations of $\mathbb{R}^3$ correspond to antipodal pairs of unit quaternions.

Sketch. $C_t(q) = tq\bar{t}$ is $\mathbb{R}$ -linear, norm-preserving, and preserves the real part, hence restricts to an isometry of $\mathrm{Im} \mathbb{H}$, of determinant $+1$ since $\mathrm{Sp}(1) \cong S^3$ is

connected and $C_1 = \text{id}$. Writing $t = \cos \theta + u \sin \theta$, one checks $C_t(u) = u$ and, on the plane $u^\perp \subset \text{Im } \mathbb{H}$, $C_t(v) = v \cos 2\theta + (uv) \sin 2\theta$: a rotation through 2θ about u . Surjectivity follows from Euler's rotation theorem, and $C_t = \text{id}$ forces t to commute with all of $\mathbb{H}$, hence to lie in the centre $\mathbb{R}$, so the kernel is $\{\pm 1\}$. Details in [Sti08]. $\square$

REMARK 2.3. The homomorphism is a smooth double covering $S^3 \rightarrow \text{SO}(3)$, so $\text{SO}(3) \cong S^3 / \{\pm 1\} = \mathbb{RP}^3$. The lens spaces $S^3 / \mathbb{Z}_{k+1}$ appear in Chapter 6 as the boundaries at infinity of the A_k spaces.

REMARK 2.4. The three left multiplications by i, j, k on $\mathbb{H} \cong \mathbb{R}^4$ are orthogonal complex structures satisfying the quaternion relations $J_1 J_2 = J_3$, and their common stabiliser in $\text{SO}(4)$, the group of quaternion-linear isometries, is $\text{Sp}(1)$.

2.2 Connections, holonomy, and Berger's theorem

Let $E \rightarrow M$ be a vector bundle over a smooth manifold M .

DEFINITION 2.5. A *connection* ∇ is a map $\nabla : \Gamma(E) \rightarrow \Gamma(T^*M \otimes E)$ satisfying

$$\begin{aligned} \nabla(s + t) &= \nabla(s) + \nabla(t), & \forall s, t \in \Gamma(E), \\ \nabla(fs) &= df \otimes s + f\nabla(s), & \forall s \in \Gamma(E), \forall f \in C^\infty(M). \end{aligned} \quad (2.4)$$

For the tangent bundle of a Riemannian manifold (M, g) , there is a canonical choice of connection called the *Levi-Civita connection* which, by the fundamental theorem of Riemannian geometry, is the unique connection on TM that is both *compatible with g* , $\nabla g = 0$, and *torsion-free*, $\nabla_X Y - \nabla_Y X = [X, Y]$ for all vector fields X, Y . We say a section s along a curve $\gamma : [0, 1] \rightarrow M$ is *parallel* if $\nabla_{\gamma'(t)} s = 0$. By ODE theory, each $s_0 \in E_{\gamma(0)}$ admits a unique parallel section $\tilde{s}$ with $\tilde{s}(0) = s_0$. Hence we can define the *parallel transport map* $P_\gamma : E_{\gamma(0)} \rightarrow E_{\gamma(1)}$. Fix $p \in M$ and consider loops γ based at p .

DEFINITION 2.6. The *holonomy group* at p is

$$\text{Hol}_p(\nabla) = \{P_\gamma : E_p \rightarrow E_p \mid \gamma \text{ is a loop based at } p\}. \quad (2.5)$$

This is a Lie subgroup of $\text{GL}(E_p)$ [Joy00, §2.2]: concatenating loops composes the transport maps, and reversing a loop inverts them. On a connected manifold the holonomy group is independent of the base point up to conjugation. An important perspective, when considering whether the holonomy group of a manifold reduces, is the following principle.

LEMMA 2.7 (Holonomy principle; see [Joy00, §2.5]). *Let T be a tensor field on M . If T is parallel, that is $\nabla T = 0$, then $\text{Hol}_p(\nabla) \subseteq \text{Stab}(T_p)$ for every p . Conversely, if $\text{Hol}_p(\nabla) \subseteq \text{Stab}(T_0)$ for some tensor T_0 at p , then T_0 extends by parallel transport to a well-defined parallel tensor field on M .*

REMARK 2.8. For the Levi-Civita connection the metric itself is parallel, so $\text{Hol}(g) \subseteq \text{O}(n)$, and $\text{Hol}(g) \subseteq \text{SO}(n)$ when M is orientable with parallel volume form. The lemma tells us that every further reduction of the holonomy group is equivalent to the existence of some additional parallel tensor: a parallel complex structure gives $\text{U}(m)$, a parallel holomorphic volume form on top of that gives $\text{SU}(m)$, and a parallel quaternionic triple gives $\text{Sp}(m)$.

DEFINITION 2.9. The *Riemann curvature tensor* of (M, g) is

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z, \quad (2.6)$$

and the *Ricci tensor* is its trace, $\text{Ric}(X, Y) = \text{tr}(Z \mapsto R(Z, X)Y)$, in local coordinates $\text{Ric}_{ij} = R^k_{ikj}$.

We say that (M, g) is *Ricci-flat* if $\text{Ric} = 0$: this is weaker than flatness $R = 0$, but stronger than the Einstein condition $\text{Ric} = \lambda g$. For a generic orientable Riemannian n -manifold we have $\text{Hol}(g) = \text{SO}(n)$ and we say (M, g) has *special holonomy* if $\text{Hol}(g)$ is a proper subgroup of $\text{SO}(n)$. It is natural to ask which holonomy groups can occur. Berger answered this in 1955, via the classification of simple Lie groups, and Simons [Sim62] later gave an intrinsic proof, noticing that the listed groups act transitively on the unit sphere of the tangent space. One notes that Berger's original list contained a further case, namely $\text{Spin}(9)$, which was later shown to only occur for symmetric spaces, see [Ale68].

THEOREM 2.10 (Berger, 1955 [Ber55]). *Let (M, g) be a simply connected Riemannian manifold of dimension n that is irreducible and non-symmetric. Then $\text{Hol}(g)$ is one of the following:*

Holonomy	Geometry	Dimension	Parallel structure	Curvature
$\text{SO}(n)$	generic	n	none	none
$\text{U}(m)$	Kähler	$2m, m \geq 2$	ω	none
$\text{SU}(m)$	Calabi-Yau	$2m, m \geq 2$	ω, Ω	$\text{Ric} = 0$
$\text{Sp}(m)$	hyperkähler	$4m, m \geq 1$	$\omega_1, \omega_2, \omega_3$	$\text{Ric} = 0$
$\text{Sp}(m) \cdot \text{Sp}(1)$	quaternionic Kähler	$4m, m \geq 2$	Φ (4-form)	$\text{Ric} = \lambda g, \lambda \neq 0$
G_2	G_2	7	φ (3-form)	$\text{Ric} = 0$
$\text{Spin}(7)$	$\text{Spin}(7)$	8	Ψ (4-form)	$\text{Ric} = 0$

CHAPTER 3

Complex Geometry

In this chapter, in the interest of not writing another whole thesis, we introduce only the complex geometry needed to frame the Calabi conjecture in Chapter 4; for further reading we refer to [Huy05], or to [Par26] for a gentler introduction.

3.1 Complex manifolds

DEFINITION 3.1. A *complex manifold* is a smooth manifold equipped with an atlas of charts to $\mathbb{C}^m$ whose transition functions are biholomorphic.

Multiplication by i in holomorphic charts gives every complex manifold a canonical endomorphism of its tangent bundle. Abstracting this structure gives the following.

DEFINITION 3.2. An *almost complex structure* on a smooth manifold M is a smooth endomorphism $J : TM \rightarrow TM$ satisfying $J^2 = -\text{Id}$.

However, not every almost complex structure arises from a complex atlas in this way. The obstruction is measured by the *Nijenhuis tensor*

$$N_J(X, Y) = [X, Y] + J[JX, Y] + J[X, JY] - [JX, JY], \quad X, Y \in \mathfrak{X}(M).$$

THEOREM 3.3 (Newlander-Nirenberg [NN57]). An almost complex structure J on M is integrable, i.e. induced by a complex structure, if and only if $N_J \equiv 0$.

REMARK 3.4. The ‘only if’ direction is immediate by computation. For the converse, note that $N_J \equiv 0$ is equivalent to involutivity, $[\Gamma(T^{0,1}), \Gamma(T^{0,1})] \subseteq \Gamma(T^{0,1})$, of the anti-holomorphic distribution. When J is real-analytic one can apply the holomorphic Frobenius theorem. The content of Theorem 3.3 is the harder C^∞ case.

3.2 Line bundles, the canonical bundle, and adjunction

DEFINITION 3.5. The *canonical bundle* of a complex manifold X of complex dimension m is the top exterior power of the holomorphic cotangent bundle,

$$K_X = \Lambda^{m,0} T^* X. \quad (3.1)$$

A *holomorphic volume form* is a nowhere-vanishing holomorphic section $\Omega \in \Gamma(K_X)$; locally $\Omega = h dz_1 \wedge \cdots \wedge dz_m$ with h holomorphic and non-vanishing.

Thus K_X is holomorphically trivial when a holomorphic volume form exists, and triviality of K_X implies $c_1(X) = 0$. It is natural to ask which complex manifolds admit a trivial canonical bundle. The following theorem gives us such examples.

THEOREM 3.6 (Adjunction for hypersurfaces). *Let $Z \subset \mathbb{CP}^n$ be a smooth hypersurface of degree d . Then*

$$K_Z \cong \mathcal{O}(d - n - 1)|_Z, \quad (3.2)$$

in particular K_Z is trivial if and only if $d = n + 1$.

Proof. See [Huy05, Prop. 2.2.17]. □

3.3 The ordinary double point

The family of Section 1.1 degenerates, at $t = 0$, to a surface with an ordinary double point, and for the purpose of this thesis we will focus on this type of singularity.

DEFINITION 3.7. An *ordinary double point* (ODP) of a complex surface is a singular point at which the surface is locally isomorphic to the quadric cone

$$Q_0 = \{z_1^2 + z_2^2 + z_3^2 = 0\} \subset \mathbb{C}^3. \quad (3.3)$$

The cone has a second description, as the simplest quotient singularity.

PROPOSITION 3.8. *The quadric cone Q_0 is isomorphic, as a complex variety, to the quotient $\mathbb{C}^2 / \{\pm 1\}$.*

Proof. The map:

$$\pi(u, v) = (u^2 - v^2, i(u^2 + v^2), 2uv), \quad (3.4)$$

has image in Q_0 since $(u^2 - v^2)^2 - (u^2 + v^2)^2 + 4u^2v^2 = 0$, and is invariant under ± 1 . The quotient is the affine variety with coordinate ring $\mathbb{C}[u, v]^{\mathbb{Z}_2}$, generated

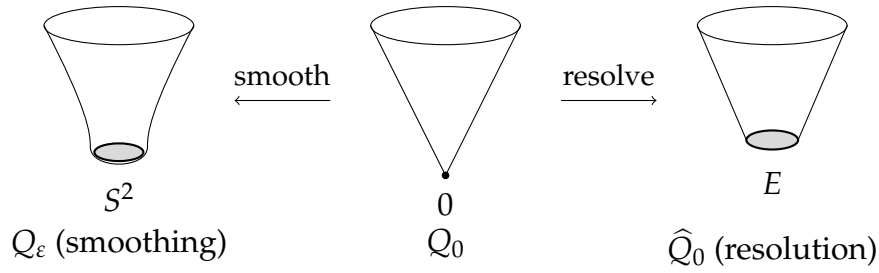


Figure 3.1: The two ways of removing an ordinary double point. Centre: the quadric cone $Q_0 \cong \mathbb{C}^2 / \{\pm 1\}$. Left: the smoothing Q_ϵ , which contains a vanishing two-sphere. Right: the resolution $\hat{Q}_0 = T^*\mathbb{CP}^1$, with exceptional curve E of self-intersection -2 .

by $A = u^2$, $B = v^2$, $C = uv$ with the relation $AB = C^2$. The change $z_1 = A - B$, $z_2 = i(A + B)$, $z_3 = 2C$ carries $\{AB = C^2\}$ to Q_0 , since $z_1^2 + z_2^2 + z_3^2 = 4(C^2 - AB)$, and π induces the isomorphism. [Dur79] $\square$

There are two ways of removing the singularity (Figure 3.1). We may *smooth* it, replacing Q_0 by $Q_\epsilon = \{z_1^2 + z_2^2 + z_3^2 = \epsilon\}$; Q_ϵ contains a two-sphere S^2 which collapses as $\epsilon \rightarrow 0$. Or we may *resolve* it by blowing up the space at the singularity. Here the singular point is replaced by a curve, the exceptional curve E , which is a copy of $\mathbb{CP}^1 \cong S^2$.

DEFINITION 3.9. The *blow-up* of $\mathbb{C}^2$ at the origin is

$$\text{Bl}_0 \mathbb{C}^2 = \{(z, \ell) \in \mathbb{C}^2 \times \mathbb{CP}^1 : z \in \ell\}, \quad (3.5)$$

together with the projection $\sigma : \text{Bl}_0 \mathbb{C}^2 \rightarrow \mathbb{C}^2$. The *exceptional divisor* $E = \sigma^{-1}(0) \cong \mathbb{CP}^1$ is contracted by σ , which is an isomorphism away from it; equivalently, $\text{Bl}_0 \mathbb{C}^2$ is the total space of the tautological bundle $\mathcal{O}_{\mathbb{CP}^1}(-1)$, with E its zero section. See [Huy05, §2.5].

Since our singularity is a quotient of $\mathbb{C}^2$, its resolution is obtained by blowing up upstairs and then dividing.

PROPOSITION 3.10. The involution $(u, v) \mapsto (-u, -v)$ of $\mathbb{C}^2$ lifts to $\text{Bl}_0 \mathbb{C}^2$, acting trivially on E and by -1 on each fibre of $\mathcal{O}_{\mathbb{CP}^1}(-1)$. The quotient

$$\hat{Q}_0 := \text{Bl}_0 \mathbb{C}^2 / \{\pm 1\} \quad (3.6)$$

is smooth, is isomorphic to the total space of $\mathcal{O}_{\mathbb{CP}^1}(-2) \cong T^*\mathbb{CP}^1$, and the induced map $\hat{Q}_0 \rightarrow \mathbb{C}^2 / \{\pm 1\} \cong Q_0$ is an isomorphism away from the exceptional curve $E \cong \mathbb{CP}^1$, which maps to the singular point: $\hat{Q}_0$ is a resolution of the ordinary double point.

Proof. The involution preserves lines through the origin, so it lifts to $\text{Bl}_0 \mathbb{C}^2 = \mathcal{O}_{\mathbb{CP}^1}(-1)$, fixing each line and acting by -1 on the fibre. Fibrewise squaring

identifies $\mathcal{O}(-1)/\{\pm 1\}$ with $\mathcal{O}(-2)$, which is $T^*\mathbb{CP}^1$ since $K_{\mathbb{CP}^1} = \mathcal{O}(-2)$; the map to Q_0 is induced by σ and Proposition 3.8, an isomorphism away from E . $\square$

These two methods produce diffeomorphic four-manifolds: the smoothing Q_ε and the resolution $T^*\mathbb{CP}^1$ are both diffeomorphic to T^*S^2 , with the vanishing two-sphere of the smoothing corresponding to the exceptional curve of the resolution. The global spaces of Chapter 8 are quotients of manifolds by finite groups, singular along the fixed points, and the right language for them is Satake's notion of an orbifold [Sat56], in the formulation of Joyce.

DEFINITION 3.11 (Joyce [Joy00, Def. 6.5.1]). An *orbifold* is a singular real manifold X of dimension n whose singularities are locally isomorphic to quotient singularities $\mathbb{R}^n/G$ for finite subgroups $G \subset \mathrm{GL}(n, \mathbb{R})$, such that if $1 \neq \gamma \in G$ then the subspace $V_\gamma \subseteq \mathbb{R}^n$ fixed by γ has $\dim V_\gamma \leq n - 2$. For each singular point $x \in X$ there is a finite subgroup $G_x \subset \mathrm{GL}(n, \mathbb{R})$, unique up to conjugation, such that open neighbourhoods of x in X and of 0 in $\mathbb{R}^n/G_x$ are homeomorphic (and, in a suitable sense, diffeomorphic). We call x an *orbifold point* of X and G_x its *orbifold group*.

Every orbifold in this thesis arises from the basic construction: if a finite group G acts smoothly and faithfully on an oriented manifold M preserving orientation, then M/G is an orbifold whose orbifold groups are the stabilisers [Joy00, Prop. 6.5.2]. Geometric structures on an orbifold (metrics, Kähler structures, hyperkähler triples) are G_x -invariant such structures in the local charts. The example to keep in mind is $T^4/\{\pm 1\}$: a flat orbifold whose sixteen singular points each have orbifold group $\{\pm 1\}$, so that each is an ordinary double point by Proposition 3.8, and the machinery of this section (resolution by $T^*\mathbb{CP}^1$) applies at every one of them. That observation is the starting point of the Kummer construction (Section 8.2).

CHAPTER 4

The Calabi Conjecture

After Berger's classification it took the best part of forty years to show that each group on the list does in fact occur. In this chapter we address the case $\text{Hol}(g) \subseteq \text{SU}(m)$, where remarkably the existence question reduces to a single scalar partial differential equation.

4.1 Kähler metrics

The setting for the Calabi conjecture lies within a subset of complex geometry known as Kähler geometry.

DEFINITION 4.1. A Riemannian metric g on an almost complex manifold (M, J) is *Hermitian* if it is compatible with J , i.e.

$$g(JX, JY) = g(X, Y), \quad \forall X, Y \in \mathfrak{X}(M). \quad (4.1)$$

In this case the *fundamental 2-form* is

$$\omega(X, Y) = g(JX, Y). \quad (4.2)$$

Compatibility makes ω alternating, and any two of (g, J, ω) determine the third.

DEFINITION 4.2. A Hermitian manifold (M, g, J) is *Kähler* if $d\omega = 0$ and ω is called its *Kähler form*.

PROPOSITION 4.3 (see [Joy00, Ch. 4]). *For a Hermitian manifold (M, g, J) with J integrable, the following are equivalent:*

- (i) $d\omega = 0$;
- (ii) $\nabla J = 0$, where ∇ is the Levi-Civita connection of g ;
- (iii) $\text{Hol}(g) \subseteq \text{U}(m)$, where $\dim_{\mathbb{C}} M = m$.

The equivalence of (ii) and (iii) is the holonomy principle (Lemma 2.7); the Kähler condition is the first non-generic entry of Berger's list.

EXAMPLE 4.4. On $\mathbb{C}^m$ the form $\omega_0 = \frac{i}{2}\partial\bar{\partial}|z|^2 = \frac{i}{2}\sum_j dz^j \wedge d\bar{z}^j$ is closed, and its metric is the flat Euclidean metric. On $\mathbb{CP}^m$ the *Fubini-Study* form $\omega_{\text{FS}} = \frac{i}{2}\partial\bar{\partial}\log(1+|w|^2)$ provides the Kähler structure. Finally, a complex submanifold $\iota : Y \hookrightarrow X$ of a Kähler manifold is Kähler in the restricted metric, since $\iota^*\omega$ is closed and positive. In particular, every smooth projective variety carries the Kähler metric $\iota^*\omega_{\text{FS}}$.

4.2 The Ricci form and the first Chern class

DEFINITION 4.5. On a Kähler manifold (M, g, J, ω) the *Ricci form* is

$$\rho(X, Y) = \text{Ric}(JX, Y). \quad (4.3)$$

Because Ric is symmetric and J -invariant on a Kähler manifold, ρ is a real $(1, 1)$ -form. In local holomorphic coordinates $\rho = -i\partial\bar{\partial}\log\det(g_{j\bar{k}})$, so ρ is closed, and less obviously its class is $[\rho] = 2\pi c_1(M) \in H^2(M, \mathbb{R})$.

REMARK 4.6. If K_M is holomorphically trivial then $c_1(M) = 0$ and ρ is exact. It does *not* follow that $\rho \equiv 0$. The vanishing of c_1 is a statement about a cohomology class whereas Ricci-flatness pertains to a particular metric.

4.3 The $\partial\bar{\partial}$ -Lemma

LEMMA 4.7 ($\partial\bar{\partial}$ -lemma [Joy00, Ch. 4]). Let (X, ω) be a compact Kähler manifold and let α be a d -exact real $(1, 1)$ -form. Then there exists a smooth real function φ on X , unique up to an additive constant, such that $\alpha = i\partial\bar{\partial}\varphi$.

More generally, a d -closed (p, q) -form is d -exact if and only if it is $\partial\bar{\partial}$ -exact.

4.4 The Calabi conjecture and its reduction to Monge-Ampère

CONJECTURE 4.8 (Calabi, 1954 [Cal54, Cal57]). Let (M, J) be a compact complex manifold and let g be a Kähler metric on M with Kähler form ω . Suppose ρ' is a real, closed $(1, 1)$ -form on M with $[\rho'] = 2\pi c_1(M)$. Then there exists a unique Kähler metric g' on M , with Kähler form ω' , such that $[\omega'] = [\omega] \in H^2(M, \mathbb{R})$ and the Ricci form of g' is ρ' .

In words: within a fixed Kähler class the Ricci form may be prescribed arbitrarily. We are interested in the $\rho' = 0$ case.

COROLLARY 4.9. *Granting Conjecture 4.8, if $c_1(M) = 0$ in $H^2(M, \mathbb{R})$, then each Kähler class on M contains a unique Ricci-flat Kähler metric.*

We now reduce Conjecture 4.8 to a scalar PDE; write $m = \dim_{\mathbb{C}} M$. Since $[\rho'] = [\rho] = 2\pi c_1(M)$, Lemma 4.7 provides a smooth real f , unique up to a constant, with $\rho - \rho' = i\partial\bar{\partial}f$. Now suppose ω' is any Kähler form and define F by $(\omega')^m = e^F \omega^m$. Comparing the local expressions for the two Ricci forms, we have

$$\rho' - \rho = -i\partial\bar{\partial}F, \quad (4.4)$$

so the metric g' has Ricci form ρ' if and only if $F = f$ up to a constant. This is an important simplification: prescribing the Ricci form, which is second order in the metric, has become prescribing the volume form, which is zeroth order in the metric. Since $[\omega']^m = [\omega]^m$, integrating $(\omega')^m = e^F \omega^m$ over M gives $\int_M e^F \omega^m = \int_M \omega^m$. This fixes the additive constant from the definition of f , which we normalise henceforth so that

$$\int_M e^f \omega^m = \int_M \omega^m. \quad (4.5)$$

Finally, $\omega' - \omega$ is an exact real $(1,1)$ -form, so Lemma 4.7 gives a smooth real function ϕ , unique once we impose $\int_M \phi dV_g = 0$, with $\omega' = \omega + i\partial\bar{\partial}\phi$. After substituting this gives the complex Monge-Ampère equation

$$(\omega + i\partial\bar{\partial}\phi)^m = e^f \omega^m \quad \text{on } M, \quad (4.6)$$

subject to $\omega + i\partial\bar{\partial}\phi$ being a positive $(1,1)$ -form. To summarise:

THEOREM 4.10 (Calabi conjecture, PDE form). *Let (M, J) be a compact complex manifold with Kähler metric g and Kähler form ω , and let f be a smooth real function on M satisfying $\int_M e^f \omega^m = \int_M \omega^m$. Then there exists a unique smooth real function ϕ on M such that*

- (i) $\omega + i\partial\bar{\partial}\phi$ is a positive $(1,1)$ -form;
- (ii) $\int_M \phi dV_g = 0$; and
- (iii) $(\omega + i\partial\bar{\partial}\phi)^m = e^f \omega^m$ on M .

4.5 The continuity method and Yau's theorem

The equation is solved by applying the continuity method. For $t \in [0, 1]$ set $f_t = tf + c_t$, the constant c_t chosen so that $\int_M e^{f_t} \omega^m = \int_M \omega^m$, and define

$$S := \left\{ t \in [0, 1] : \exists \phi \text{ satisfying (i), (ii) and } (\omega + i\partial\bar{\partial}\phi)^m = e^{f_t} \omega^m \right\}. \quad (4.7)$$

Since $f_0 = 0$, the function $\phi \equiv 0$ solves the equation at $t = 0$, so $S \neq \emptyset$. Since $[0, 1]$ is connected it suffices to show S is open and closed. The analysis is best seen in local holomorphic coordinates where we write $g_{\alpha\bar{\beta}}$ for the metric and $\phi_{\alpha\bar{\beta}} = \partial^2 \phi / \partial z^\alpha \partial \bar{z}^\beta$. The equation becomes

$$\det(g_{\alpha\bar{\beta}} + \phi_{\alpha\bar{\beta}}) = e^{f_t} \det(g_{\alpha\bar{\beta}}), \quad (g_{\alpha\bar{\beta}} + \phi_{\alpha\bar{\beta}}) > 0, \quad (4.8)$$

as in [Joy00, §5.1]: this is a scalar, second-order, fully nonlinear equation, elliptic at every ϕ satisfying the positivity constraint, since its linearisation there is the Laplacian Δ_ϕ of the metric $\omega_\phi := \omega + i\partial\bar{\partial}\phi$.

Openness (Calabi). At a solution, Δ_ϕ is an isomorphism $C_0^{2,\alpha} \rightarrow C_0^{0,\alpha}$ between Hölder spaces of mean-zero functions and applying the implicit function theorem for Banach spaces gives existence for nearby t . Uniqueness follows by Calabi's integration-by-parts argument: if ϕ_1, ϕ_2 both solve the equation then $0 = \int_M (\phi_1 - \phi_2)(\omega_1^m - \omega_2^m)$ with $\omega_i = \omega + i\partial\bar{\partial}\phi_i$, and expanding $\omega_1^m - \omega_2^m = i\partial\bar{\partial}(\phi_1 - \phi_2) \wedge \sum_{a+b=m-1} \omega_1^a \wedge \omega_2^b$ and integrating by parts leaves a sum of non-negative gradient terms, forcing $\phi_1 - \phi_2$ to be constant zero by (ii).

REMARK 4.11. The implicit function theorem produces $\phi \in C^{2,\alpha}$, while S asks for ϕ smooth. This is resolved by the fact that a $C^{2,\alpha}$ solution is automatically smooth by elliptic regularity. Δ_ϕ then has $C^{0,\alpha}$ coefficients and Schauder estimates bootstrap to all orders.

Closedness (Yau). Given $t_n \rightarrow t$ in S with solutions ϕ_n , one needs a priori bounds on ϕ_n , uniform in n , strong enough to pass to a limit solving the equation at t . See [Yau78] or [Joy00, Ch. 5] for a full treatment.

THEOREM 4.12 (Yau [Yau78]). *The Calabi conjecture holds.*

4.6 Calabi-Yau manifolds and examples

DEFINITION 4.13. A *Calabi-Yau manifold* is a compact Kähler manifold M of complex dimension m with K_M holomorphically trivial, equipped with a Ricci-flat

Kähler metric g supplied by Theorem 4.12. The trivialising holomorphic volume form is then parallel, so $\text{Hol}(g) \subseteq \text{SU}(m)$ by Lemma 2.7.

For the projective examples below, K_M is trivial by adjunction (Theorem 3.6).

EXAMPLE 4.14. By Theorem 3.6: cubic curves in $\mathbb{CP}^2$ (tori, with flat metrics), quartic K3 surfaces in $\mathbb{CP}^3$, and quintic threefolds in $\mathbb{CP}^4$.

CHAPTER 5

Hyperkähler Metrics from Closed Two-Forms

Following Theorem 4.12, one would like to know how such metrics behave. This leads us to Chapter 6, but first we prove a fundamental lemma used in said construction. The proposition appears in [Don06] and [Don12, §5].

PROPOSITION 5.1. *Let M^4 be a smooth four-manifold oriented by a volume form V , and let $\omega_1, \omega_2, \omega_3$ be two-forms which satisfy*

$$\omega_i \wedge \omega_j = \delta_{ij} V. \quad (5.1)$$

Then there is a unique Riemannian metric g on M with $\text{vol}_g = V/2$ such that $\text{span}\{\omega_1, \omega_2, \omega_3\} = \Lambda_g^+$ pointwise, and for which $(\omega_1, \omega_2, \omega_3)$ is an orthogonal frame with $|\omega_i|_g^2 = 2$. Moreover, if $d\omega_i = 0$ for $i = 1, 2, 3$, then g is hyperkähler and

$$\text{Hol}(g) \subseteq \text{Sp}(1) \cong \text{SU}(2). \quad (5.2)$$

The proof has three parts: on each tangent space the algebraic condition alone determines a unique Euclidean structure (Sections 5.1 and 5.2), the pointwise metrics vary smoothly (Section 5.3), and closedness forces the triple to be parallel, so the holonomy principle applies (Section 5.4).

5.1 Pointwise existence

LEMMA 5.2. *Let E be an oriented four-dimensional $\mathbb{R}$ -vector space and $\omega_i \in \Lambda^2 E^*$ satisfying $\omega_i \wedge \omega_j = \delta_{ij} V$, where V is a positively oriented volume element. Then there exists a unique Euclidean structure g on E such that $\Lambda_g^+ = \text{span}(\{\omega_i\})$ and $\text{vol}_g = V/2$. The ω_i are then g -orthogonal with $|\omega_i|_g^2 = 2$.*

In the existence proof, ω_2 and ω_3 produce a complex structure I on E , and ω_1 then pairs with I , as a Kähler form pairs with a complex structure, to produce the metric.

Proof of existence.

Step 1: Decomposability of Θ .

Define $\Theta = \omega_2 + i\omega_3$. We claim that Θ is decomposable, i.e. a wedge product of two complex one-forms. Expanding $\Theta \wedge \Theta$ and using the hypotheses,

$$\Theta \wedge \Theta = \omega_2^2 - \omega_3^2 + 2i\omega_2 \wedge \omega_3 = 0, \quad \Theta \wedge \bar{\Theta} = \omega_2^2 + \omega_3^2 = 2V \neq 0. \quad (5.3)$$

Decomposability then follows from a general fact: any non-zero two-form τ on a four-dimensional vector space with $\tau \wedge \tau = 0$ is decomposable. To see this, pick v with $\alpha := \iota_v \tau \neq 0$; contracting $\tau \wedge \tau = 0$ with v gives $2\alpha \wedge \tau = 0$. Set $f_1 := \alpha$, extend to a basis $\{f_i\}$ of E^* , and write $\tau = t^{ij}f_i \wedge f_j$. The relation $f_1 \wedge \tau = 0$ forces $t^{ij} = 0$ for $i, j \geq 2$, so $\tau = \alpha \wedge \beta$ with $\beta = t^{1j}f_j$. Hence $\Theta = \alpha \wedge \beta$ for some $\alpha, \beta \in E^* \otimes \mathbb{C}$.

Step 2: The complex structure I .

Using this we define a complex structure on E . Let

$$K := \{X \in E \otimes_{\mathbb{R}} \mathbb{C} : \iota_X \Theta = 0\} = \{\alpha = \beta = 0\}, \quad (5.4)$$

the common kernel of the functionals α and β . These are linearly independent (otherwise Θ would vanish), so K is a complex subspace of dimension 2. Let $\bar{K}$ be defined analogously for $\bar{\Theta}$. If $X \in K \cap \bar{K}$, then contracting $\Theta \wedge \bar{\Theta} = 2V$ with X gives $\iota_X V = 0$, so $X = 0$. Counting dimensions gives us

$$E \otimes \mathbb{C} = K \oplus \bar{K}. \quad (5.5)$$

We now define $I \in \text{End}(E \otimes \mathbb{C})$ to be $+i$ on $\bar{K}$ and $-i$ on K . Since I commutes with conjugation, it is the complexification of a genuine complex structure on E . By construction $T^{0,1} = K$, thus $\Lambda^{1,0} = \text{span}_{\mathbb{C}}(\alpha, \beta)$ and Θ spans the line $\Lambda^{2,0}$. (The choice of sign is made so that α and β vanish on $E^{0,1}$, i.e. so that Θ is a $(2,0)$ -form.)

Step 3: ω_1 has type $(1,1)$.

Decompose $\omega_1 = \omega_1^{2,0} + \omega_1^{1,1} + \omega_1^{0,2}$, and note that $\omega_1^{2,0} = \overline{\omega_1^{0,2}}$ since ω_1 is real. Wedging ω_1 with Θ and comparing types, the hypothesis $\omega_1 \wedge \Theta = 0$ leaves only $\omega_1^{0,2} \wedge \Theta = 0$. Since the pairing $\Lambda^{2,0} \times \Lambda^{0,2} \rightarrow \Lambda^{2,2}$ is nondegenerate, both the $(2,0)$ - and $(0,2)$ -components vanish and $\omega_1 = \omega_1^{1,1}$. Equivalently, since I acts by -1 on $\Lambda^{2,0} \oplus \Lambda^{0,2}$ and by $+1$ on $\Lambda^{1,1}$,

$$\omega_1(IX, IY) = \omega_1(X, Y). \quad (5.6)$$

Step 4: Construct the metric.

We can now write down the metric. Let

$$\tilde{g} := \omega_1(\cdot, I\cdot). \quad (5.7)$$

This is symmetric, by the type $(1, 1)$ property of ω_1 :

$$\tilde{g}(Y, X) = \omega_1(Y, IX) = \omega_1(IY, I^2X) = \omega_1(X, IY) = \tilde{g}(X, Y). \quad (5.8)$$

It is nondegenerate, since $\omega_1^2 = V \neq 0$, and it is Hermitian: $\tilde{g}(I\cdot, I\cdot) = \tilde{g}(\cdot, \cdot)$. What we do not yet know is that $\tilde{g}$ is definite, and indeed it need not be positive: this is why we introduce the sign ε below. Define

$$h(X, Y) := \tilde{g}(X, Y) + i\omega_1(X, Y), \quad (5.9)$$

a Hermitian form on (E, I) . Diagonalising h gives us complex coordinates z_1, z_2 adapted to I and $\lambda_1, \lambda_2 \in \mathbb{R} \setminus \{0\}$ such that

$$\tilde{g} = \lambda_1 |dz_1|^2 + \lambda_2 |dz_2|^2, \quad \omega_1 = \frac{i}{2}(\lambda_1 dz_1 \wedge d\bar{z}_1 + \lambda_2 dz_2 \wedge d\bar{z}_2), \quad (5.10)$$

and $\Theta = c dz_1 \wedge dz_2$ for some $c \neq 0$, since Θ spans $\Lambda^{2,0}$. Let $dV_0 = dx_1 dy_1 dx_2 dy_2$ denote the orientation. Using $\frac{i}{2} dz \wedge d\bar{z} = dx \wedge dy$ and $dz_1 \wedge dz_2 \wedge d\bar{z}_1 \wedge d\bar{z}_2 = 4 dV_0$, we compute

$$\omega_1^2 = 2\lambda_1\lambda_2 dV_0, \quad \Theta \wedge \bar{\Theta} = 4|c|^2 dV_0, \quad (5.11)$$

so $V = 2|c|^2 dV_0$, and comparing with $\omega_1^2 = V$ gives

$$\lambda_1\lambda_2 = |c|^2 > 0. \quad (5.12)$$

The two eigenvalues have the same sign, so $\tilde{g}$ is definite. Let $\varepsilon := \text{sign}(\lambda_1)$; then $g := \varepsilon \tilde{g}$ is positive definite.

Step 5: $\Lambda_g^+ = \text{span}(\{\omega_i\})$ and the normalisations.

Rescale to unitary coordinates:

$$v_a := \sqrt{|\lambda_a|} z_a, \quad (5.13)$$

and write

$$v_1 = u_0 + iu_1, \quad v_2 = u_2 + iu_3, \quad e^a := du_a. \quad (5.14)$$

Then

$$g = \sum_a (e^a)^2, \quad (5.15)$$

the coframe (e^0, e^1, e^2, e^3) is oriented, and

$$\varepsilon \omega_1 = e^{01} + e^{23} =: \sigma_1, \quad (5.16)$$

$$\Theta = e^{i\varphi} dv_1 \wedge dv_2 = e^{i\varphi}(\sigma_2 + i\sigma_3), \quad \sigma_2 = e^{02} + e^{31}, \quad \sigma_3 = e^{03} + e^{12}, \quad (5.17)$$

where $e^{i\varphi} := c/|c|$. Thus $\omega_2 = \operatorname{Re} \Theta$ and $\omega_3 = \operatorname{Im} \Theta$ are obtained from the pair (σ_2, σ_3) by a rotation through the angle φ . Since $(\sigma_1, \sigma_2, \sigma_3)$ is the standard frame of Λ_g^+ , we deduce

$$\Lambda_g^+ = \operatorname{span}(\omega_1, \omega_2, \omega_3), \quad \langle \omega_i, \omega_j \rangle_g = 2 \delta_{ij}, \quad (5.18)$$

and

$$V = \omega_1^2 = \sigma_1^2 = 2e^{0123} = 2dV_g, \quad \text{so} \quad dV_g = V/2. \quad (5.19)$$

This establishes pointwise existence of g . $\square$

5.2 Pointwise uniqueness

Let h be *any* Euclidean structure on E such that each ω_i is h -self-dual (self-duality taken with the V -orientation) and $dV_h = V/2$. We want to show that $h = g$. The hypotheses already determine the inner products of the ω_i : for $\sigma, \tau \in \Lambda_h^+$,

$$\sigma \wedge \tau = \sigma \wedge *_h \tau = \langle \sigma, \tau \rangle_h dV_h, \quad (5.20)$$

so

$$\omega_i \wedge \omega_j = \delta_{ij} V \implies \langle \omega_i, \omega_j \rangle_h = 2 \delta_{ij}. \quad (5.21)$$

LEMMA 5.3. *Let $\sigma \in \Lambda^2 E^*$ be self-dual or anti-self-dual for h , and let $A_\sigma \in \operatorname{End}(E)$ be the associated skew endomorphism, i.e. $\sigma(X, Y) = h(A_\sigma X, Y)$. Then*

$$A_\sigma^2 = -\frac{1}{2} |\sigma|_h^2 \operatorname{Id}. \quad (5.22)$$

In words: a self-dual form, viewed as an endomorphism, is a complex structure up to scale. We first prove a normal form.

LEMMA 5.4. *Let $\sigma \in \Lambda_h^+ \setminus \{0\}$. Then there exists an oriented h -orthonormal coframe in which*

$$\sigma = a (e^{01} + e^{23}), \quad \text{where} \quad a^2 = \frac{|\sigma|_h^2}{2}. \quad (5.23)$$

Proof. We have $\sigma(X, Y) = h(A_\sigma X, Y)$, i.e. $A_\sigma X = (\iota_X \sigma)^\sharp$. Choose e_0 with $|e_0| = 1$ and $A_\sigma e_0 \neq 0$, and let

$$e_1 := \frac{A_\sigma e_0}{|A_\sigma e_0|}. \quad (5.24)$$

Then $e_1 \perp e_0$, since skewness of A_σ gives

$$h(A_\sigma e_0, e_0) = -h(e_0, A_\sigma e_0) = -h(A_\sigma e_0, e_0) = 0. \quad (5.25)$$

Complete to an oriented h -orthonormal basis (e_0, e_1, e_2, e_3) . Write $\sigma = \sigma_{ij} e^{ij}$ (sum over $i < j$), so that

$$\iota_{e_0} \sigma = \sigma_{01} e^1 + \sigma_{02} e^2 + \sigma_{03} e^3. \quad (5.26)$$

On the other hand $\iota_{e_0} \sigma = (A_\sigma e_0)^\flat = |A_\sigma e_0| e^1$ by our choice of e_1 ; it follows that $\sigma_{02} = \sigma_{03} = 0$. Now self-duality pairs up the coefficients as $\sigma_{01} = \sigma_{23}$, $\sigma_{02} = \sigma_{31}$, $\sigma_{03} = \sigma_{12}$, so

$$\sigma = \sigma_{01} (e^{01} + e^{23}), \quad a = \sigma_{01} = |A_\sigma e_0| > 0, \quad (5.27)$$

and $|\sigma|_h^2 = \sigma_{01}^2 + \sigma_{23}^2 = 2a^2$. $\square$

Proof of Lemma 5.3. Suppose first that σ is self-dual. The anti-self-dual case follows by reversing the orientation. Let u be any unit vector and let $\{e_i\}$ be an oriented h -orthonormal basis with $e_0 = u$. Then

$$|A_\sigma u|^2 = |\iota_u \sigma|^2 = \sigma_{01}^2 + \sigma_{02}^2 + \sigma_{03}^2, \quad (5.28)$$

and by self-duality ($\sigma_{01} = \sigma_{23}$, etc.) this is half of $|\sigma|_h^2 = \sum_{i < j} \sigma_{ij}^2$:

$$|A_\sigma u|^2 = \frac{1}{2} |\sigma|_h^2. \quad (5.29)$$

The right-hand side is basis-independent and u was arbitrary so $|A_\sigma u|$ is the same in every direction. Since A_σ is skew we have A_σ^2 is symmetric and

$$h(A_\sigma^2 u, u) = -|A_\sigma u|^2 = -\frac{1}{2} |\sigma|_h^2 \quad (5.30)$$

is constant on the unit sphere. A symmetric bilinear form which is constant on unit vectors (here, constant equal to $-\frac{1}{2} |\sigma|_h^2$) is by polarisation a multiple of h and therefore

$$A_\sigma^2 = -\frac{1}{2} |\sigma|_h^2 \text{Id}. \quad (5.31)$$

$\square$

COROLLARY 5.5 (quaternion relations). For $\sigma, \tau \in \Lambda_h^+$,

$$A_\sigma A_\tau + A_\tau A_\sigma = -\langle \sigma, \tau \rangle_h \text{Id}. \quad (5.32)$$

In particular, the $J_i := A_{\omega_i}$ satisfy

$$J_i^2 = -\text{Id}, \quad J_i J_j = -J_j J_i \quad (i \neq j). \quad (5.33)$$

Proof. The map $\sigma \mapsto A_\sigma$ is linear and Λ_h^+ is a linear subspace, so Lemma 5.3 applies to $\sigma + \tau \in \Lambda_h^+$:

$$A_{\sigma+\tau}^2 = -\frac{1}{2}|\sigma + \tau|_h^2 \text{Id}. \quad (5.34)$$

Expanding both sides and subtracting the diagonal terms A_σ^2 and A_τ^2 gives the identity. The relations for the J_i follow since $\langle \omega_i, \omega_j \rangle_h = 2\delta_{ij}$. The triple reconstructs the quaternionic structure of Section 2.1. $\square$

Proof of uniqueness.

(i) J_2J_3 recovers I .

Set $\tilde{I} := J_2J_3$. We want to show that $\tilde{I} = I$. Note first that $\tilde{I}^2 = J_2J_3J_2J_3 = -\text{Id}$ by Corollary 5.5, so $\tilde{I}$ is a complex structure, and it suffices to identify its $(-i)$ -eigenspace. For $X \in E \otimes \mathbb{C}$,

$$\iota_X \Theta = 0 \iff h(J_2X + iJ_3X, \cdot) = 0 \iff J_2X = -iJ_3X \iff J_2J_3X = -iX, \quad (5.35)$$

where the last step applies J_3 and uses the quaternion relations. It follows that $\tilde{I}$ has $(-i)$ -eigenspace $K = \ker \Theta$ and, by conjugating, $(+i)$ -eigenspace $\bar{K}$. These are the defining properties of I , so $J_2J_3 = \tilde{I} = I$.

(ii) Comparing $\tilde{g}$ with h .

Consider the triple product $A := J_1J_2J_3$. Since J_1 anticommutes with J_2 and J_3 , it commutes with J_2J_3 , giving $A^2 = J_1^2(J_2J_3)^2 = \text{Id}$ and $A^\top = (-J_3)(-J_2)(-J_1) = -J_3J_2J_1 = A$. So A is an h -symmetric involution. Now

$$\tilde{g}(X, Y) = \omega_1(X, IY) = h(J_1X, J_2J_3Y) = -h(X, AY). \quad (5.36)$$

On the (± 1) -eigenspaces of A , the form $-h(\cdot, A \cdot)$ equals $\mp h$. If both eigenspaces were nonzero $\tilde{g}$ would be indefinite which contradicts Step 4 of the existence proof. Thus

$$A = \eta \text{Id}, \quad \eta \in \{\pm 1\}, \quad (5.37)$$

and $\tilde{g} = -\eta h$. Since $h > 0$, the sign of $\tilde{g}$ is $\varepsilon = -\eta$, and it follows

$$h = -\eta \tilde{g} = \varepsilon \tilde{g} = g. \quad (5.38)$$

$\square$

5.3 Smoothness of g

Applying Lemma 5.2 on each tangent space produces a metric g_x for every $x \in M^4$. It remains to check that $x \mapsto g_x$ is smooth. Every object in the construction can be put together into a smooth bundle map. Let

$$\Phi : TM \otimes \mathbb{C} \longrightarrow T^*M \otimes \mathbb{C}, \quad \Phi(v) = \iota_v \Theta. \quad (5.39)$$

Then Φ is smooth and has rank 2 at every point, its kernel at x being the 2-dimensional space K_x . A smooth bundle map of constant rank has smooth kernel, so $K = \ker \Phi$ is a smooth rank-2 subbundle, and so is $\bar{K}$. Then

$$TM \otimes \mathbb{C} = K \oplus \bar{K}, \quad (5.40)$$

and with respect to a local frame $\{k_1, k_2, \bar{k}_1, \bar{k}_2\}$ adapted to this splitting, I is a matrix with smooth entries. Since ω_1 and I are smooth, and the sign ε is locally constant (by continuity of $\tilde{g}(Z, Z)$ for a local non-vanishing vector field Z), we conclude that $x \mapsto g_x = \varepsilon \omega_1(\cdot, I \cdot)$ is smooth.

5.4 Holonomy reduction: $\text{Hol}(g) \subseteq \text{Sp}(1)$

Suppose now that $d\omega_i = 0$ for $i = 1, 2, 3$. The plan is to show that closedness forces the ω_i to be *parallel*, after which the holonomy principle finishes the proof. Before starting, relabel $\omega_1 := \varepsilon \omega_1$, so that $g = \omega_1(\cdot, I \cdot)$. The computation below is that of [Don12, §5.1], carried out on the self-dual bundle Λ^+ .

Step 1: Λ^+ is ∇ -invariant.

Let $\{e^i\}$ be a local g -orthonormal coframe and let

$$\sigma_1 = e^{01} + e^{23}, \quad \sigma_2 = e^{02} + e^{31}, \quad \sigma_3 = e^{03} + e^{12} \quad (5.41)$$

be the associated standard frame of Λ^+ . Write $\nabla e^a = -\Gamma_{ab} \otimes e^b$ with $\Gamma_{ab} = -\Gamma_{ba}$ the connection one-forms of the Levi-Civita connection. Differentiating,

$$\begin{aligned} \nabla e^{01} &= \Gamma_{02} \otimes e^{12} + \Gamma_{03} \otimes e^{13} - \Gamma_{12} \otimes e^{02} - \Gamma_{13} \otimes e^{03}, \\ \nabla e^{23} &= \Gamma_{02} \otimes e^{03} + \Gamma_{12} \otimes e^{13} - \Gamma_{03} \otimes e^{02} - \Gamma_{13} \otimes e^{12}. \end{aligned} \quad (5.42)$$

Setting

$$a_1 = \Gamma_{01} + \Gamma_{23}, \quad a_2 = \Gamma_{02} + \Gamma_{31}, \quad a_3 = \Gamma_{03} + \Gamma_{12}, \quad (5.43)$$

the two lines add up to

$$\nabla \sigma_1 = a_2 \otimes \sigma_3 - a_3 \otimes \sigma_2, \quad (5.44)$$

and similarly $\nabla \sigma_2, \nabla \sigma_3 \in \Gamma(T^*M \otimes \Lambda^+)$. Hence ∇ preserves Λ^+ , inducing a connection on the rank-3 bundle Λ^+ .

Step 2: The connection matrix in the frame (ω_i) .

By Step 1, there exist one-forms a_{ij} such that

$$\nabla \omega_i = a_{ij} \otimes \omega_j. \quad (5.45)$$

Differentiating $\langle \omega_i, \omega_j \rangle_g = 2\delta_{ij}$ gives $a_{ij} = -a_{ji}$. We may therefore write

$$a_{ij} = \varepsilon_{ijk} \gamma_k \quad (5.46)$$

for three one-forms $\gamma_1, \gamma_2, \gamma_3$. It remains to show the γ_k vanish.

Step 3: The exterior derivatives.

Since ∇ is torsion-free, the exterior derivative is the skew-symmetrisation of ∇ :

$$d\beta = e^k \wedge \nabla_{E_k} \beta \quad \text{for any form } \beta, \text{ frame } \{E_k\} \text{ and its dual coframe } \{e^k\}. \quad (5.47)$$

Applying this to $\nabla \omega_i = a_{ij} \otimes \omega_j$ gives $d\omega_i = a_{ij} \wedge \omega_j$ which written out is

$$\begin{aligned} d\omega_1 &= \gamma_3 \wedge \omega_2 - \gamma_2 \wedge \omega_3, \\ d\omega_2 &= -\gamma_3 \wedge \omega_1 + \gamma_1 \wedge \omega_3, \\ d\omega_3 &= \gamma_2 \wedge \omega_1 - \gamma_1 \wedge \omega_2. \end{aligned} \quad (5.48)$$

Step 4: Applying the Hodge star.

Taking the star of the first equation

$$*d\omega_1 = *(\gamma_3 \wedge \omega_2) - *(\gamma_2 \wedge \omega_3). \quad (5.49)$$

We use the identity $*(\beta \wedge \gamma) = \iota_{\gamma^\sharp}(*\beta)$ valid for γ a one-form. Since ω_2 is self-dual,

$$*(\gamma_3 \wedge \omega_2) = \iota_{\gamma_3^\sharp} \omega_2 = g(J_2 \gamma_3^\sharp, \cdot) = (J_2 \gamma_3^\sharp)^\flat =: J_2 \gamma_3, \quad (5.50)$$

where we abbreviate $(J_i \gamma^\sharp)^\flat$ by $J_i \gamma$ again for a one-form γ . Therefore

$$*d\omega_1 = J_2 \gamma_3 - J_3 \gamma_2, \quad \text{and similarly for } *d\omega_2, *d\omega_3. \quad (5.51)$$

Step 5: Solving.

The system of Step 4 is linear in the γ_k with coefficients in the quaternionic endomorphisms J_i , one finds

$$\gamma_1 = \frac{1}{2} \left[J_2(*d\omega_3) - J_3(*d\omega_2) - *d\omega_1 \right], \quad (5.52)$$

and its cyclic analogues. So if $d\omega_i = 0$ for all i , then

$$\gamma_1 = \gamma_2 = \gamma_3 = 0 \implies a_{ij} = 0 \implies \nabla \omega_i = 0 \implies \nabla J_i = 0. \quad (5.53)$$

By the holonomy principle (Lemma 2.7),

$$\text{Hol}(g) \subseteq \text{Stab}(\omega_1, \omega_2, \omega_3) = \text{Sp}(1), \quad (5.54)$$

where the stabiliser in $\text{SO}(4)$ of the triple $(\omega_1, \omega_2, \omega_3)$, equivalently of the quaternionic triple (J_1, J_2, J_3) , is the group of quaternion-linear isometries $\text{Sp}(1)$, as noted in Section 2.1. This completes the proof of Proposition 5.1. $\square$

REMARK 5.6 (Hypersymplectic triples and Donaldson's conjecture). Proposition 5.1 has 2 hypotheses: the pointwise algebraic condition $\omega_i \wedge \omega_j = \delta_{ij}V$ and the differential condition $d\omega_i = 0$. This leads to a natural weakening: a triple of closed two-forms $(\omega_1, \omega_2, \omega_3)$ on a four-manifold is called *hypersymplectic* if the symmetric matrix Q defined by $\omega_i \wedge \omega_j = 2Q_{ij}\mu$, for a fixed volume form μ , is positive definite at every point [Don06]. A positive definite triple is the case $Q \equiv \text{Id}$, and Donaldson conjectured that, on a *compact* four-manifold, this is the only case up to deformation: every hypersymplectic structure can be deformed, through cohomologous hypersymplectic structures, to a hyperkähler triple [Don06]. One way of tackling this problem is through Bryant's Laplacian flow, a flow on closed G_2 -structures. A hypersymplectic triple on M defines a closed G_2 -structure $\varphi = dt_{123} + \sum_i dt_i \wedge \omega_i$ on $M \times T^3$ (compare Section 8.3), which is torsion-free precisely when the triple is hyperkähler, and Bryant's Laplacian flow $\partial_t \varphi = \Delta_\varphi \varphi$ [Bry06] preserves closedness. This induces the *hypersymplectic flow* of Fine and Yao. Fine and Yao proved that this flow exists for as long as its scalar curvature remains bounded [FY18]. The conjecture remains open.

5.5 The curvature operator as a block matrix

We record here an aside on the curvature of four-manifolds: in dimension four the curvature decomposes into blocks along $\Lambda^2 = \Lambda^+ \oplus \Lambda^-$. Let $\mathcal{R}$ be the *curvature operator*, the symmetric endomorphism of $\Lambda^2 = \Lambda^2 T^*M$ determined by

$$\langle \mathcal{R}(X^\flat \wedge Y^\flat), U^\flat \wedge V^\flat \rangle = \langle R(X, Y)V, U \rangle. \quad (5.55)$$

THEOREM 5.7 (Singer-Thorpe [ST69]; see also [Bes87, Ch. 1.H]). *With respect to $\Lambda^2 = \Lambda^+ \oplus \Lambda^-$,*

$$\mathcal{R} = \begin{pmatrix} W^+ + \frac{s}{12} \text{Id} & \text{Ric}_0 \\ \text{Ric}_0^\top & W^- + \frac{s}{12} \text{Id} \end{pmatrix}, \quad (5.56)$$

where s is the scalar curvature, the off-diagonal block is the trace-free Ricci curvature viewed as a map $\Lambda^- \rightarrow \Lambda^+$, and the Weyl curvatures $W^\pm \in S_0^2(\Lambda^\pm)$ are the trace-free parts of the diagonal blocks.

REMARK 5.8. In the hyperkähler case, the top row $(W^+ + \frac{s}{12} \text{Id} \mid \text{Ric}_0)$ vanishes, leaving W^- as the only non-vanishing curvature element. Indeed the top row is the curvature of the connection induced on Λ^+ . A hyperkähler triple trivialises Λ^+ by parallel sections, so this connection is flat and the row vanishes. For a full proof see [Dun10, Thm 9.3.3].

CHAPTER 6

The Gibbons-Hawking Construction

Proposition 5.1 reduces the construction of hyperkähler four-manifolds to producing positive definite triples. In this chapter we present the construction of Gibbons and Hawking [GH78], which produces such triples from positive harmonic functions on open subsets of $\mathbb{R}^3$. Once more we follow [Don12, §5].

6.1 The construction

Chapter 2 introduced connections on vector bundles. The construction below uses the complementary formalism of connection one-forms on circle bundles, which we now fix.

DEFINITION 6.1. A *principal S^1 -bundle* over a manifold U is a manifold P with a free action of $S^1 = \mathbb{R}/\mathbb{Z}$ and an invariant map $\pi : P \rightarrow U$ inducing a diffeomorphism $P/S^1 \rightarrow U$, locally trivial in the sense that every point of U has a neighbourhood V with an equivariant diffeomorphism $\pi^{-1}(V) \cong V \times S^1$ over V . Write ξ for the vector field generating the action. A *connection* on P is an S^1 -invariant one-form $\alpha \in \Omega^1(P)$ with $\alpha(\xi) = 1$. Its *curvature* is the two-form $d\alpha$, which is the pullback of a two-form on the base U .

Now let f be a positive harmonic function on an open set $U \subseteq \mathbb{R}^3$. We write $f_i = \partial f / \partial x_i$ and $f_{ij} = \partial^2 f / \partial x_i \partial x_j$ for its partial derivatives, and $*$ for the Euclidean Hodge star of U , so that

$$*df = f_1 dx_2 \wedge dx_3 + f_2 dx_3 \wedge dx_1 + f_3 dx_1 \wedge dx_2. \quad (6.1)$$

Since f is harmonic, $*df$ is closed. Suppose there is a principal S^1 -bundle $\pi : P \rightarrow U$ with a connection one-form α whose curvature is $-*df$:

$$d\alpha = -\pi^*(* df). \quad (6.2)$$

On the total space P define the metric and the triple of two-forms

$$g = f(dx_1^2 + dx_2^2 + dx_3^2) + f^{-1}\alpha^2, \quad \omega_i = \alpha \wedge dx_i + f dx_j \wedge dx_k, \quad (6.3)$$

for (i, j, k) a cyclic permutation of $(1, 2, 3)$, where we suppress pullbacks. We observe that the circles collapse where f blows up.

THEOREM 6.2 (Gibbons-Hawking [GH78]). *The forms $\omega_1, \omega_2, \omega_3$ of (6.3) produce a closed definite triple on P : setting $V := 2f\alpha \wedge dx_1 \wedge dx_2 \wedge dx_3$,*

$$\omega_i \wedge \omega_j = \delta_{ij} V, \quad d\omega_i = 0. \quad (6.4)$$

The hyperkähler metric that Proposition 5.1 associates to this triple is $g = f dx^2 + f^{-1}\alpha^2$.

Proof. For the wedge relations: $\omega_i \wedge \omega_i = 2\alpha \wedge dx_i \wedge f dx_j \wedge dx_k = V$, while for $i \neq j$ every term of $\omega_i \wedge \omega_j$ contains a repeated one-form factor and vanishes. For closedness, equation (6.2) gives

$$d\omega_i = d\alpha \wedge dx_i + df \wedge dx_j \wedge dx_k = -(*df) \wedge dx_i + df \wedge dx_j \wedge dx_k = 0. \quad (6.5)$$

Finally,

$$(e^0, e^1, e^2, e^3) := (f^{-1/2}\alpha, f^{1/2} dx_1, f^{1/2} dx_2, f^{1/2} dx_3) \quad (6.6)$$

is an orthonormal coframe for g , in which

$$\omega_i = e^0 \wedge e^i + e^j \wedge e^k \quad (6.7)$$

is the standard self-dual frame of Section 5.4. So the ω_i span Λ_g^+ and are orthogonal with $|\omega_i|_g^2 = 2$, and $dV_g = e^{0123} = f\alpha \wedge dx_1 \wedge dx_2 \wedge dx_3 = V/2$. $\square$

EXAMPLE 6.3 (Flat space). Take $U = \mathbb{R}^3 \setminus \{0\}$ and $f(x) = \frac{1}{4\pi|x|}$. The resulting four-manifold is flat $\mathbb{R}^4 \setminus \{0\}$, filling in smoothly over the origin, with the Hopf map as circle fibration. Writing $\mathbb{R}^4 = \mathbb{C}^2 \ni (u_1, u_2)$, the circle fibration is the Hopf map $x_3 = \pi(|u_1|^2 - |u_2|^2)$, $x_1 + ix_2 = 2\pi \bar{u}_1 u_2$, under which $|x| = \pi|u|^2$.

6.2 The Eguchi-Hanson metric

The first non-trivial output of the construction is the two-pole metric.

DEFINITION 6.4. The *Eguchi-Hanson space* is the Gibbons-Hawking space of the two-pole harmonic function

$$f(x) = \frac{1}{4\pi} \left(\frac{1}{|x - p_1|} + \frac{1}{|x - p_2|} \right), \quad p_1 \neq p_2 \in \mathbb{R}^3. \quad (6.8)$$

Topologically the Eguchi-Hanson space is the cotangent bundle T^*S^2 . It contains a distinguished *central two-sphere* Σ over the segment $[p_1, p_2]$, of self-intersection $\Sigma \cdot \Sigma = -2$, matching the resolution of $\mathbb{C}^2/\{\pm 1\}$ of Section 3.3. At

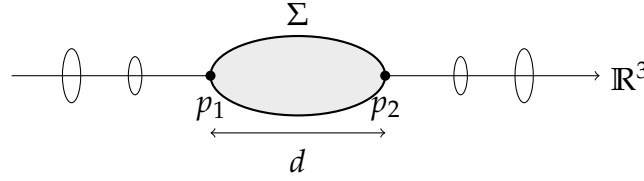


Figure 6.1: The Eguchi-Hanson space in its Gibbons-Hawking presentation. The circle fibre collapses at the two poles and lengthens like $\sqrt{|x|}$ at infinity. Over the segment connecting the poles the fibres form a central two-sphere Σ .

infinity the space is asymptotic to the cone over $\mathbb{RP}^3 = S^3/\mathbb{Z}_2$ (Figure 6.1). The same metric has a second, older presentation [EH78]. Fix Euler angles (θ, ϕ, ψ) on $SU(2) \cong S^3$ and the coframe

$$\begin{aligned} \eta_1 &= \cos \psi d\theta + \sin \psi \sin \theta d\phi, & \eta_2 &= -\sin \psi d\theta + \cos \psi \sin \theta d\phi, \\ \eta_3 &= d\psi + \cos \theta d\phi, \end{aligned} \quad (6.9)$$

which satisfies $d\eta_i = -\eta_j \wedge \eta_k$ for (i, j, k) cyclic, with $\eta_1^2 + \eta_2^2 = d\theta^2 + \sin^2 \theta d\phi^2$; see [EGH80]. In terms of this coframe, a radial coordinate $r > a$, and the function

$$h(r) = 1 - \frac{a^4}{r^4}, \quad (6.10)$$

the metric is

$$g_a = h(r)^{-1} dr^2 + \frac{r^2}{4} (\eta_1^2 + \eta_2^2) + \frac{r^2}{4} h(r) \eta_3^2, \quad (6.11)$$

with the central two-sphere the set $\{r = a\}$. These two presentations describe the same metric. The explicit change of coordinates is given in [Pra79], see also [Hit79].

6.3 ALE spaces and the A_k family

The Eguchi-Hanson space is complete, and at infinity it approaches the flat cone $\mathbb{C}^2/\{\pm 1\}$ to leading order $O(r^{-4})$ by reading off $h(r)$. This motivates the definition of a class of spaces known as asymptotically locally Euclidean (ALE) spaces. Below, G is a finite subgroup of $\mathrm{Sp}(1)$ acting on $\mathbb{H} \cong \mathbb{R}^4$, $(\tilde{J}_1, \tilde{J}_2, \tilde{J}_3, \tilde{g})$ is the Euclidean hyperkähler structure descended to $\mathbb{H}/G$, and $r : \mathbb{H}/G \rightarrow [0, \infty)$ is the radius function.

DEFINITION 6.5 (ALE hyperkähler four-manifold; [Joy00, Def. 7.2.1]). Let G be a finite subgroup of $\mathrm{Sp}(1)$. A hyperkähler four-manifold (X, J_1, J_2, J_3, g) is *asymptotically locally Euclidean* (ALE) and *asymptotic to $\mathbb{H}/G$* if there exist a compact

subset $S \subset X$ and a map $\pi : X \setminus S \rightarrow \mathbb{H}/G$ which is a diffeomorphism onto $\{x \in \mathbb{H}/G : r(x) > R\}$ for some $R > 0$, such that

$$\tilde{\nabla}^k(\pi_*(g) - \tilde{g}) = O(r^{-4-k}) \quad \text{and} \quad \tilde{\nabla}^k(\pi_*(J_j) - \tilde{J}_j) = O(r^{-4-k}) \quad (6.12)$$

as $r \rightarrow \infty$, for $j = 1, 2, 3$ and all $k \geq 0$, where $\tilde{\nabla}$ is the Levi-Civita connection of $\tilde{g}$.

REMARK 6.6. Flat $\mathbb{H} = \mathbb{R}^4$ is trivially ALE asymptotic to $\mathbb{H}/\{1\}$, and the Eguchi-Hanson space is ALE asymptotic to $\mathbb{H}/\{\pm 1\}$. This definition asks for decay of the whole hyperkähler structure relative to a fixed identification of the end with the flat cone. This “closeness”, in the sense that ends almost match is crucial in the gluing constructions of Chapter 8.

DEFINITION 6.7. The A_k space is the Gibbons-Hawking space of

$$f(x) = \frac{1}{4\pi} \sum_{a=1}^{k+1} \frac{1}{|x - p_a|}, \quad (6.13)$$

for distinct points $p_1, \dots, p_{k+1} \in \mathbb{R}^3$.

Thus A_0 is flat $\mathbb{R}^4$ and A_1 is Eguchi-Hanson. The space A_k is asymptotic to a cone over the lens space $S^3/\mathbb{Z}_{k+1}$ and it contains a chain of k two-spheres $\Sigma_1, \dots, \Sigma_k$ lying over the segments $[p_a, p_{a+1}]$.

REMARK 6.8 (completeness and asymptotics). The A_k space is a complete hyperkähler four-manifold, ALE asymptotic to $\mathbb{H}/\mathbb{Z}_{k+1}$ in the sense of Definition 6.5.

REMARK 6.9 (Kronheimer’s classification). The Gibbons-Hawking construction covers only cyclic quotients $\mathbb{H}/\mathbb{Z}_{k+1}$. The full classification of hyperkähler ALE spaces is due to Kronheimer [Kro89a, Kro89b], via the hyperkähler quotient construction [HKLR87, §3] and the McKay correspondence between the finite subgroups $G \subset \text{SU}(2)$ and the Dynkin diagrams A_r, D_r, E_6, E_7, E_8 . See [Joy00, Thm. 7.2.3] for the precise statement.

REMARK 6.10 (ALF spaces and gravitational instantons). Since we only need f to be harmonic, it is natural to consider another family of functions, namely:

$$f = c + \sum_{i=1}^n \frac{1}{4\pi|x - p_i|}, \quad c > 0, \quad (6.14)$$

which gives the *Taub-NUT metric* ($n = 1$) [Haw77] and in general the *multi-Taub-NUT* family [GH78]. The fibre circles of length $f^{-1/2}$ tend to the finite length $c^{-1/2}$ instead of growing like $\sqrt{|x|}$, so the end is a circle bundle over the complement of a ball in $\mathbb{R}^3$ with fibres of bounded length, and volume grows like ρ^3 in place of ρ^4 .

These spaces are called *asymptotically locally flat* (ALF) rather than ALE. Both classes fit into the theory of *gravitational instantons*, complete non-compact hyperkähler four-manifolds of finite energy $\int_X |\text{Rm}|^2 dV_g < \infty$ [SZ24]. The different asymptotic geometries are organised by the volume growth $\text{vol } B(\rho) \sim \rho^\nu$: ALE, ALF, ALG, ALH have $\nu = 4, 3, 2, 1$ respectively [CC21].

The Curvature of Gibbons-Hawking Metrics

Following [Don12, §5], we compute the curvature of an *arbitrary* Gibbons-Hawking metric from its potential: all the non-vanishing curvature of a hyperkähler four-manifold lies in the anti-self-dual bundle, and the connection-form computation of Section 5.4 gives the full Riemann tensor as a symmetric trace-free 3×3 matrix in two derivatives of f (Theorem 7.2).

7.1 The curvature tensor in Gibbons-Hawking coordinates

The anti-self-dual frame and its connection forms

Throughout, $g = f dx^2 + f^{-1} \alpha^2$ is a Gibbons-Hawking metric (6.3) and (i, j, k) is a cyclic permutation of $(1, 2, 3)$. Alongside the closed triple ω_i , introduce the anti-self-dual two-forms

$$\theta_i := \alpha \wedge dx_i - f dx_j \wedge dx_k = e^0 \wedge e^i - e^j \wedge e^k, \quad (7.1)$$

an orthogonal frame of Λ^- with $|\theta_i|^2 = 2$. The full Riemann tensor is encoded in the matrix (W_{ij}) defined by

$$\mathcal{R} \theta_i = \sum_j W_{ij} \theta_j, \quad (7.2)$$

symmetric (pair symmetry of $\mathcal{R}$) and trace-free. Throughout this chapter $|W|$ denotes the operator norm of (W_{ij}) , equivalently of the endomorphism $\mathcal{R}|_{\Lambda^-}$ in the orthogonal frame $\theta_i / \sqrt{2}$. Since $\mathcal{R}$ vanishes on Λ^+ , this is comparable to the full curvature norm $|\text{Rm}|_g$ up to a dimensional constant.

LEMMA 7.1 (connection forms of Λ^-). *For any Gibbons-Hawking metric,*

$$d\theta_i = -2f_i dx_1 \wedge dx_2 \wedge dx_3, \quad (7.3)$$

and there is a unique triple of one-forms (T_1, T_2, T_3) satisfying the structure equations

$$d\theta_i = T_k \wedge \theta_j - T_j \wedge \theta_k, \quad (i, j, k) \text{ cyclic}, \quad (7.4)$$

namely

$$T_i = -\frac{f_i}{f^2} \alpha + \frac{1}{f} (f_j dx_k - f_k dx_j). \quad (7.5)$$

These are the connection forms of the Levi-Civita connection on Λ^- in the frame (θ_i) : $\nabla \theta_i = a_{ij} \otimes \theta_j$ with $a_{ij} = \varepsilon_{ijk} T_k$.

Proof. For (7.3): $d\theta_i = d\alpha \wedge dx_i - df \wedge dx_j \wedge dx_k$, and with $d\alpha = - * df$ each term equals $-f_i dx_1 \wedge dx_2 \wedge dx_3$. Writing $T_i = p_i \alpha + \sum_l q_{il} dx_l$ and using the fact that $-2 T_i = * \left[d\theta_i - \theta_j \wedge (* d\theta_k) + \theta_k \wedge (* d\theta_j) \right]$ by comparing coefficients one calculates the T_i 's. $\square$

THEOREM 7.2 (the curvature tensor; Donaldson [Don12, §5]). *Define the curvature two-forms $F_i := dT_i + T_j \wedge T_k$, (i, j, k) cyclic. Then*

$$F_i = \sum_j W_{ij} \theta_j, \quad W_{ij} = \text{trace-free part of } M_{ij} := \frac{f_{ij}}{f^2} - 3 \frac{f_i f_j}{f^3}, \quad (7.6)$$

and (W_{ij}) is the matrix (7.2): the trace-free part of (M_{ij}) is the Riemann tensor.

Proof. The F_i are the curvature forms of the connection of Lemma 7.1: for a connection written in a frame as $\nabla \theta_i = a_{ij} \otimes \theta_j$, the curvature acts by

$$R^{\Lambda^2}(X, Y) \theta_i = \sum_j (da_{ij} - a_{il} \wedge a_{lj})(X, Y) \theta_j, \quad (7.7)$$

and substituting $a_{ij} = \varepsilon_{ijk} T_k$ gives $R^{\Lambda^2}(X, Y) \theta_i = \sum_j \varepsilon_{ijk} F_k(X, Y) \theta_j$ with $F_k = dT_k + T_i \wedge T_j$. It remains to compute F_1 and read off the matrix. From (7.5), expand dT_1 and $T_2 \wedge T_3$. Collecting coefficients, the six components of $F_1 = dT_1 + T_2 \wedge T_3$ are

$$\begin{aligned} \alpha \wedge dx_1 : & \quad \frac{f_{11}}{f^2} - \frac{2f_1^2 - f_2^2 - f_3^2}{f^3}, & dx_2 \wedge dx_3 : & \quad \frac{f_{22} + f_{33}}{f} + \frac{2f_1^2 - f_2^2 - f_3^2}{f^2}, \\ \alpha \wedge dx_2 : & \quad M_{12}, & dx_3 \wedge dx_1 : & \quad -f M_{12}, \\ \alpha \wedge dx_3 : & \quad M_{13}, & dx_1 \wedge dx_2 : & \quad -f M_{13}. \end{aligned} \quad (7.8)$$

In the basis $\theta_j = \alpha \wedge dx_j - f dx_k \wedge dx_l$ the off-diagonal terms are $M_{12} \theta_2 + M_{13} \theta_3$. For the diagonal term we use that f is harmonic: $f_{22} + f_{33} = -f_{11}$ makes the pair a multiple of θ_1 , with coefficient

$$\frac{f_{11}}{f^2} - \frac{2f_1^2 - f_2^2 - f_3^2}{f^3} = M_{11} + \frac{|\nabla f|^2}{f^3} = M_{11} - \frac{1}{3} \operatorname{tr} M, \quad (7.9)$$

since $\operatorname{tr} M = \frac{\Delta f}{f^2} - 3 \frac{|\nabla f|^2}{f^3} = -3 \frac{|\nabla f|^2}{f^3}$. So $F_1 = \sum_j (M_0)_{1j} \theta_j$ with M_0 the trace-free part of M . The other terms follow identically. $\square$

REMARK 7.3 (Hessian form, and the flat examples). Since $\partial_i \partial_j (f^{-2}) = -2f^{-3} f_{ij} + 6f^{-4} f_i f_j$,

$$W = -\frac{f}{2} \left[\operatorname{Hess} (f^{-2}) \right]_0, \quad (7.10)$$

$[\cdot]_0$ denoting the trace-free part. So $W \equiv 0$ forces $f^{-2} = c|x - x_0|^2 + b$, and harmonicity forces $f = \operatorname{const}$ (flat $\mathbb{R}^3 \times S^1$) or $f = c'/|x - x_0|$ (flat $\mathbb{R}^4$).

7.2 The Eguchi-Hanson case

PROPOSITION 7.4 (exact curvature of Eguchi-Hanson). *For the potential of Definition 6.4, with $r_a = |x - p_a|$ and $d = |p_1 - p_2|$,*

$$W = \frac{12\pi (r_1 r_2)^2}{(r_1 + r_2)^3} \left[w w^\top \right]_0, \quad w := \frac{x - p_1}{r_1^2} - \frac{x - p_2}{r_2^2}, \quad |w| = \frac{d}{r_1 r_2}, \quad (7.11)$$

with eigenvalues

$$(2, -1, -1) \cdot \frac{4\pi d^2}{(r_1 + r_2)^3}. \quad (7.12)$$

Consequently

$$|W| = \frac{8\pi d^2}{(r_1 + r_2)^3}. \quad (7.13)$$

In the geodesic distance ρ ,

$$|W| \sim \frac{8d^2}{\pi^2} \rho^{-6} \quad \text{as } \rho \rightarrow \infty. \quad (7.14)$$

Proof.

Step 1: Reduction.

By Theorem 7.2, W is the trace-free part of $M = f^{-3}(f f_{ij} - 3 f_i f_j)$. The trace-free projection disregards multiples of Id as a result it suffices to compute $f f_{ij} - 3 f_i f_j$ modulo Id . Write $4\pi f = r_1^{-1} + r_2^{-1}$.

Step 2: Derivatives of a single pole.

For $r = |x - p|$ we have $\partial_i r = (x - p)_i / r$, hence

$$\begin{aligned}\nabla \frac{1}{r} &= -\frac{x - p}{r^3}, \\ \partial_i \partial_j \frac{1}{r} &= \frac{3(x - p)_i (x - p)_j}{r^5} - \frac{\delta_{ij}}{r^3} \equiv \frac{3(x - p)_i (x - p)_j}{r^5} \pmod{\text{Id}},\end{aligned}\tag{7.15}$$

so, summing over the two poles, modulo Id,

$$\begin{aligned}4\pi \nabla f &= -\frac{x - p_1}{r_1^3} - \frac{x - p_2}{r_2^3}, \\ 4\pi \text{Hess } f &\equiv \frac{3(x - p_1)(x - p_1)^\top}{r_1^5} + \frac{3(x - p_2)(x - p_2)^\top}{r_2^5}.\end{aligned}\tag{7.16}$$

Step 3: The quadratic combination.

Multiply out $16\pi^2(f \text{Hess } f - 3\nabla f \nabla f^\top)$. The square term $(x - p_1)(x - p_1)^\top$ appears in $f \text{Hess } f$ with coefficient $(\frac{1}{r_1} + \frac{1}{r_2})\frac{3}{r_1^5}$ and in $-3\nabla f \nabla f^\top$ with coefficient $-\frac{3}{r_1^6}$, leaving

$$\left(\frac{1}{r_1} + \frac{1}{r_2}\right)\frac{3}{r_1^5} - \frac{3}{r_1^6} = \frac{3}{r_1^5 r_2}:\tag{7.17}$$

symmetrically the second square term carries $\frac{3}{r_2^5 r_1}$. The cross terms come only from the gradient part, namely

$$-\frac{3}{r_1^3 r_2^3} \left[(x - p_1)(x - p_2)^\top + (x - p_2)(x - p_1)^\top \right].\tag{7.18}$$

Step 4: Completing the square.

Taking out common factor $3/r_1 r_2$ from the three terms leaves

$$\begin{aligned}\frac{(x - p_1)(x - p_1)^\top}{r_1^4} - \frac{(x - p_1)(x - p_2)^\top + (x - p_2)(x - p_1)^\top}{r_1^2 r_2^2} \\ + \frac{(x - p_2)(x - p_2)^\top}{r_2^4} = w w^\top,\end{aligned}\tag{7.19}$$

the expansion of the square of $w = \frac{x - p_1}{r_1^2} - \frac{x - p_2}{r_2^2} = \nabla \log(r_1/r_2)$. Hence

$$16\pi^2(f f_{ij} - 3 f_i f_j) \equiv \frac{3}{r_1 r_2} w w^\top \pmod{\text{Id}},\tag{7.20}$$

and:

$$|w|^2 = \frac{r_1^2 + r_2^2 - 2(x - p_1) \cdot (x - p_2)}{r_1^2 r_2^2} = \frac{|(x - p_1) - (x - p_2)|^2}{r_1^2 r_2^2} = \frac{d^2}{r_1^2 r_2^2}. \quad (7.21)$$

Step 5: Restoring the prefactor.

From $4\pi f = (r_1 + r_2)/r_1 r_2$ we get $f^{-3} = 64\pi^3 (r_1 r_2)^3 / (r_1 + r_2)^3$, so

$$\begin{aligned} W &= f^{-3} \cdot \frac{1}{16\pi^2} \cdot \frac{3}{r_1 r_2} [w w^\top]_0 = \frac{64\pi^3 (r_1 r_2)^3}{16\pi^2 (r_1 + r_2)^3} \cdot \frac{3}{r_1 r_2} [w w^\top]_0 \\ &= \frac{12\pi (r_1 r_2)^2}{(r_1 + r_2)^3} [w w^\top]_0, \end{aligned} \quad (7.22)$$

which is (7.11).

Step 6: Eigenvalues.

$[w w^\top]_0 = w w^\top - \frac{1}{3}|w|^2 \text{Id}$ has eigenvalue $\frac{2}{3}|w|^2$ in the direction of w and $-\frac{1}{3}|w|^2$ on the plane $w^\perp$, this gives the eigenvalues $(2, -1, -1) \cdot 4\pi d^2 / (r_1 + r_2)^3$.

Step 7: Decay at infinity.

From the eigenvalues, we have $|W| = 8\pi d^2 / (r_1 + r_2)^3$. At infinity $r_1 + r_2 = 2|x| (1 + O(|x|^{-1}))$, so $|W| \sim \pi d^2 / |x|^3$. Now by substituting the geodesic distance $\rho \approx \sqrt{2|x|/\pi}$, i.e. $|x|^3 = \pi^3 \rho^6 / 8$,

$$|W| \sim \frac{8d^2}{\pi^2} \rho^{-6}. \quad (7.23)$$

□

REMARK 7.5 (geodesic distance). A curve γ over the path $x(t) \subset \mathbb{R}^3$ has length $\int \sqrt{f|\dot{x}|^2 + f^{-1}\alpha(\dot{\gamma})^2} dt$. Since $g \geq f dx^2$ and $f(y) = \frac{1}{2\pi|y|} (1 + O(|y|^{-1}))$, every curve to x has length at least $\int^{|x|} \frac{ds}{\sqrt{2\pi s}} (1 + O(s^{-1}))$, while the horizontal lift of the straight ray to x , on which $\alpha(\dot{\gamma}) = 0$, attains this. Hence, in the flat coordinates,

$$\rho(x) = \sqrt{2|x|/\pi} + O(1). \quad (7.24)$$

REMARK 7.6 (constancy on ellipsoids). By Step 7 of the proof, $|W| = 8\pi d^2 / (r_1 + r_2)^3$ depends only on $r_1 + r_2$. The curvature norm is constant on the confocal ellipsoids $\{r_1 + r_2 = \text{const}\}$ with foci at the poles.

7.3 The curvature energy as a flux

For a compact Ricci-flat four-manifold the Chern-Gauss-Bonnet theorem makes the energy $\int |\text{Rm}|^2 dV_g$ of Remark 6.10 a topological invariant [Che44], and for an

ALE space the same holds with a correction from the end [Nak90]. We show that for a Gibbons-Hawking metric the Euler density is the divergence of a vector field on $U \subseteq \mathbb{R}^3$ built from f and its first two derivatives (Theorem 7.8). For a 3×3 matrix A write $|A|_F^2 := \sum_{i,j} A_{ij}^2$, and let $|\text{Rm}|_g^2 := R_{abcd}R^{abcd}$ in any orthonormal frame. Since the fibre circles have period one and $dV_g = f \alpha \wedge dx_1 \wedge dx_2 \wedge dx_3$ (proof of Theorem 6.2), for any S^1 -invariant function Φ ,

$$\int_P \Phi dV_g = \int_U f \Phi d^3x. \quad (7.25)$$

LEMMA 7.7. *For a Gibbons-Hawking metric, $|\text{Rm}|_g^2 = 4|W|_F^2$, and the Chern-Gauss-Bonnet integrand is $\frac{1}{8\pi^2}|W|_F^2 dV_g$.*

Proof. In an orthonormal coframe the e^{ab} , $a < b$, are an orthonormal basis of Λ^2 in which the curvature operator has entries $\pm R_{abcd}$ so its Frobenius norm squared is $\sum_{a<b, c<d} R_{abcd}^2 = \frac{1}{4}|\text{Rm}|_g^2$. This norm is independent of the orthonormal basis, and since $\mathcal{R}$ vanishes on Λ^+ and has matrix (W_{ij}) in the orthonormal basis $\theta_i/\sqrt{2}$ of Λ^- (7.2) it equals $|W|_F^2$. The Chern-Gauss-Bonnet integrand of a compact oriented four-manifold is $\frac{1}{8\pi^2}(|W^+|_F^2 + |W^-|_F^2 + \frac{s^2}{24} - \frac{1}{2}|\text{Ric}_0|^2) dV_g$, [Bes87, 6.31], and for a hyperkähler metric W^- is the only term that does not vanish. $\square$

The computation that follows is cleanest in terms of $u := f^{-2}$, recall by Remark 7.3 we have $W = -\frac{f}{2}[\text{Hess } u]_0$. Write

$$G := \nabla u, \quad H := \text{Hess } u, \quad L := \Delta u = \text{tr } H. \quad (7.26)$$

Since $f = u^{-1/2}$, $\Delta f = -\frac{1}{2}u^{-3/2}L + \frac{3}{4}u^{-5/2}|G|^2$, by the harmonicity of f it follows

$$u L = \frac{3}{2}|G|^2. \quad (7.27)$$

THEOREM 7.8 (The integrand of the curvature energy is a divergence). *Let f be a positive harmonic function on an open set $U \subseteq \mathbb{R}^3$, let W be the curvature matrix of its Gibbons-Hawking metric, and let $u = f^{-2}$. Then, on U ,*

$$f|W|_F^2 = \text{div } V, \quad V := \frac{1}{4}u^{-3/2}\left(\text{Hess } u - \frac{1}{2}(\Delta u)\text{Id}\right)\nabla u. \quad (7.28)$$

In terms of f , $V = f^{-1}W\nabla f + \frac{1}{2}f^{-4}|\nabla f|^2\nabla f$.

Proof.

Step 1: The density in terms of u .

From $W = -\frac{1}{2}u^{-1/2}[H]_0$ and $|[H]_0|_F^2 = |H|_F^2 - \frac{1}{3}L^2$, then (7.27),

$$f|W|_F^2 = \frac{1}{4}u^{-3/2}\left(|H|_F^2 - \frac{1}{3}L^2\right) = \frac{1}{4}u^{-3/2}|H|_F^2 - \frac{3}{16}u^{-7/2}|G|^4. \quad (7.29)$$

Step 2: Divergence terms

We use $\operatorname{div}(HG) = \partial_i(\partial_i \partial_j u \partial_j u) = \nabla L \cdot G + |H|_F^2$, $\operatorname{div} G = L$, $\nabla |G|^2 = 2HG$, and the gradient of (7.27), $u \nabla L + L G = 3HG$, which with (7.27) again gives $u \nabla L \cdot G = 3G^\top HG - \frac{3}{2}|G|^4/u$. Then

$$\begin{aligned} \operatorname{div}(u^{-3/2}HG) &= u^{-3/2}(|H|_F^2 + \nabla L \cdot G) - \frac{3}{2}u^{-5/2}G^\top HG \\ &= u^{-3/2}|H|_F^2 + \frac{3}{2}u^{-5/2}G^\top HG - \frac{3}{2}u^{-7/2}|G|^4, \\ \operatorname{div}(u^{-5/2}|G|^2G) &= -\frac{5}{2}u^{-7/2}|G|^4 + 2u^{-5/2}G^\top HG + u^{-5/2}|G|^2L \\ &= 2u^{-5/2}G^\top HG - u^{-7/2}|G|^4. \end{aligned} \quad (7.30)$$

Step 3: Combining terms

Subtracting $\frac{3}{4}$ of the second identity in (7.30) from the first removes the terms in $G^\top HG$:

$$\operatorname{div}\left(u^{-3/2}HG - \frac{3}{4}u^{-5/2}|G|^2G\right) = u^{-3/2}|H|_F^2 - \frac{3}{4}u^{-7/2}|G|^4, \quad (7.31)$$

which is four times (7.29). By (7.27), $\frac{3}{4}u^{-5/2}|G|^2 = \frac{1}{2}u^{-3/2}L$, so the vector field is $u^{-3/2}(H - \frac{1}{2}L \operatorname{Id})G$ and dividing by 4 gives (7.28).

Step 4: In terms of f .

From $u = f^{-2}$: $G = -2f^{-3}\nabla f$, $H = -2f^{-1}M$ with M the matrix of Theorem 7.2, $L = -2f^{-1}\operatorname{tr} M = 6f^{-4}|\nabla f|^2$ and $u^{-3/2} = f^3$, so $V = f^{-1}M\nabla f + \frac{3}{2}f^{-4}|\nabla f|^2\nabla f$; substituting $M = W - f^{-3}|\nabla f|^2 \operatorname{Id}$ gives the stated form. $\square$

THEOREM 7.9 (energy of the A_k spaces). *Let X be the A_k space of $N = k + 1$ distinct poles. Then*

$$\int_X |W|_F^2 dV_g = 8\pi^2\left(N - \frac{1}{N}\right), \quad \text{i.e.} \quad \frac{1}{32\pi^2} \int_X |\operatorname{Rm}|_g^2 dV_g = N - \frac{1}{N}. \quad (7.32)$$

Proof.

Step 1: The flux of a perturbed model.

Suppose that near a sphere $\{|y| = s\}$, $y := x - q$, we have

$$u = c^2 |y|^2 (1 + \eta), \quad c > 0, \quad |y|^m |\partial^m \eta| = O(\delta) \quad (m = 0, 1, 2), \quad (7.33)$$

for some δ ; for $\eta = 0$ this is $f = 1/(c|y|)$. Then $G = 2c^2 y(1 + O(\delta))$, $H = 2c^2 \text{Id} + O(c^2 \delta)$, so $L = 6c^2(1 + O(\delta))$, $H - \frac{1}{2}L \text{Id} = -c^2 \text{Id} + O(c^2 \delta)$ and $u^{-3/2} = c^{-3}|y|^{-3}(1 + O(\delta))$. Hence

$$\begin{aligned} V &= -\frac{c}{2} \frac{y}{|y|^3} (1 + O(\delta)), \\ \oint_{\{|y|=s\}} \left\langle V, \frac{y}{|y|} \right\rangle dA &= -\frac{c}{2} \cdot \frac{4\pi s^2}{s^2} (1 + O(\delta)) = -2\pi c + O(\delta). \end{aligned} \quad (7.34)$$

Step 2: Near a pole.

Fix $p = p_a$ and, as in Section 6.2, write $f = \frac{1}{4\pi r} + h$ with $r = |x - p|$ and h harmonic and smooth near p . Then

$$u = f^{-2} = 16\pi^2 r^2 (1 + 4\pi r h)^{-2}, \quad (7.35)$$

which is (7.33) with $q = p$, $c = 4\pi$ and $\eta = (1 + 4\pi r h)^{-2} - 1 = O(r)$ since $\partial^m(rh) = O(r^{1-m})$. The derivative bounds hold with $\delta = \varepsilon$ on $S_\varepsilon(p)$. So the flux of V through $S_\varepsilon(p)$, outward from p , is $-8\pi^2 + O(\varepsilon)$.

Step 3: At infinity.

With $r_a = |x - p_a|$,

$$f = \frac{N}{4\pi|x|} (1 + e), \quad e = \frac{1}{N} \sum_a \left(\frac{|x|}{r_a} - 1 \right), \quad (7.36)$$

and expanding each $|x|/r_a = (1 - 2 \frac{p_a \cdot x}{|x|^2} + \frac{|p_a|^2}{|x|^2})^{-1/2}$ in homogeneous functions of x of degrees $-1, -2, \dots$ gives $|x|^m \partial^m e = O(|x|^{-1})$. So $u = \frac{16\pi^2}{N^2} |x|^2 (1 + e)^{-2}$ is (7.33) with $q = 0$, $c = 4\pi/N$ and $\delta = R^{-1}$ on S_R . The outward flux of V through S_R is $-8\pi^2/N + O(R^{-1})$.

Step 4: Adding the terms

For $\varepsilon > 0$ small and R large put $\Omega_{\varepsilon,R} := B_R(0) \setminus \bigcup_a \overline{B_\varepsilon(p_a)}$, on which f is smooth and positive. By Theorem 7.8 and the divergence theorem, the outward normal of $\Omega_{\varepsilon,R}$ on $S_\varepsilon(p_a)$ pointing *towards* p_a ,

$$\begin{aligned} \int_{\Omega_{\varepsilon,R}} f |W|_F^2 d^3x &= \left(-\frac{8\pi^2}{N} + O(R^{-1}) \right) - \sum_{a=1}^N \left(-8\pi^2 + O(\varepsilon) \right) \\ &= 8\pi^2 \left(N - \frac{1}{N} \right) + O(\varepsilon) + O(R^{-1}). \end{aligned} \quad (7.37)$$

The integrand is non-negative, so letting $\varepsilon \rightarrow 0$ and $R \rightarrow \infty$ (monotone convergence) the integral over $\mathbb{R}^3 \setminus \{p_a\}$ is $8\pi^2(N - 1/N)$. (7.25) gives (7.32), with the second form by Lemma 7.7. $\square$

REMARK 7.10 (the vector field is a gradient). Since f is harmonic, $\Delta(f^{-1}) = 2|\nabla f|^2/f^3$, and substituting this into (7.28) shows that the vector field is a gradient:

$$V = \frac{1}{4} \nabla(\Delta f^{-1}), \quad \text{equivalently} \quad f |W|_F^2 = \frac{1}{4} \Delta \Delta(f^{-1}). \quad (7.38)$$

The curvature energy density of a Gibbons-Hawking metric is thus a multiple of the bi-Laplacian of $1/f$. In this form the identity appears in [NS96, eq. (32)] and in [GW00, Rmk. 2.4].

Degenerations and Gluing Constructions

We now show the importance of the Eguchi-Hanson metric: first in the degenerating quartic family of Chapter 1, then in the Kummer and Joyce gluing constructions.

8.1 Degeneration of K3 metrics

To make the picture of Section 1.1 precise, let $X_t \subset \mathbb{CP}^3$ be a family of quartic surfaces, smooth for $t \neq 0$, with X_0 carrying a single ordinary double point p . Each smooth X_t carries Ricci-flat Kähler metrics (Example 4.14); X_0 carries an *orbifold* Ricci-flat Kähler metric g_0 , by the orbifold version of Yau's theorem [Kob85], approximated near p by the flat cone metric on $\mathbb{C}^2/\{\pm 1\}$. The Eguchi-Hanson space with central two-sphere at scale a approaches the same cone, with correction $O((a/r)^4)$ (see (6.11)): the ends match, and the cut-off procedure of Section 8.2 produces a pre-glued metric, approximately Ricci-flat with error supported on the gluing annulus $a \ll r \ll 1$. The modelling statement is then: for small $t \neq 0$ the Ricci-flat metric on X_t in the appropriate class is close, in weighted norms, to the pre-glued metric.

8.2 The Kummer construction: K3 from $T^4/\{\pm 1\}$

On $T^4 = \mathbb{R}^4/\mathbb{Z}^4$ the involution $u \mapsto -u$ preserves the flat hyperkähler triple, with 16 fixed points at the half-lattice points: $X_0 = T^4/\{\pm 1\}$ is a flat orbifold with 16 ordinary double points (Section 3.3). Resolving each as in Proposition 3.10 yields the *Kummer surface*, a K3 surface [Joy00, §7.3]. The Kummer construction describes Yau's metrics on it, in classes making the 16 exceptional curves small.

REMARK 8.1 (Resolution versus smoothing). This looks different from the degeneration of Section 1.1: there the Kähler class is fixed and the complex structure varies (the double point is smoothed); here the complex structure is fixed and the

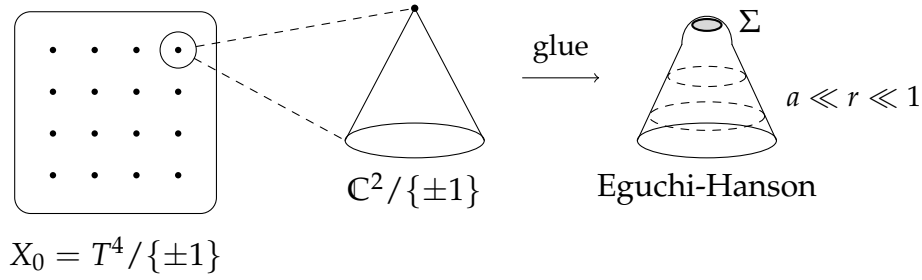


Figure 8.1: The Kummer construction. Each of the 16 double points of $T^4/\{\pm 1\}$ has a neighbourhood isometric to a ball in the cone $\mathbb{C}^2/\{\pm 1\}$; an Eguchi-Hanson space with central two-sphere at scale a has the same end, so the two metrics are interpolated on the annulus $a \ll r \ll 1$, where they differ by $O((a/r)^4)$.

Kähler class varies (the double points are resolved). The two are equivalent under a rotation of the complex structure.

Fix a small $a > 0$, away from the singular points keep the flat metric, and at each one insert an Eguchi-Hanson space with central two-sphere at scale a (Figure 8.1). On the overlap annulus $a \ll r \ll 1$ the flat metric is the cone metric, and Eguchi-Hanson differs from it by $O((a/r)^4)$. By interpolating with cut-offs we have a metric g_a on the Kummer surface, hyperkähler except on the 16 annuli, with error $O(a^4)$. The correction term is a Kähler potential: ω^2 must agree with the volume form $\frac{1}{2} \Theta \wedge \bar{\Theta}$ (the Monge-Ampère identity of a hyperkähler triple: $\omega_i \wedge \omega_j = \delta_{ij} V$ gives $\omega_1^2 = \frac{1}{2} \Theta \wedge \bar{\Theta}$ for $\Theta = \omega_2 + i\omega_3$). For g_a the two sides differ by $1 + \eta$, $\eta = O(a^4)$ supported on the annuli, and one seeks φ with

$$(\omega_a + i\partial\bar{\partial}\varphi)^2 = (1 + \eta) \omega_a^2, \quad \text{that is,} \quad \Delta_a \varphi + N(\partial\bar{\partial}\varphi) = \eta, \quad (8.1)$$

the Monge-Ampère equation (4.8) with small right-hand side, Δ_a the Laplacian of g_a and N collecting quadratic order terms.

8.3 Joyce's compact G_2 -manifolds

Until Joyce's work [Joy96] no compact manifold with holonomy exactly G_2 was known. His construction is thematically similar to that of Kummer. First we cover the language of G_2 -structures, following [Joy00, Ch. 10].

G_2 -structures

DEFINITION 8.2 ([Joy00, Def. 10.1.1]). On $\mathbb{R}^7$, with $dx_{ij\dots l} := dx_i \wedge dx_j \wedge \dots \wedge dx_l$, define the three-form

$$\varphi_0 = dx_{123} + dx_{145} + dx_{167} + dx_{246} - dx_{257} - dx_{347} - dx_{356}. \quad (8.2)$$

Its stabiliser in $GL(7, \mathbb{R})$ is the exceptional Lie group G_2 (equivalently $\text{Aut}(\mathcal{O})$): compact, connected, simply connected, of dimension 14, fixing also the Euclidean metric g_0 , the orientation, and the four-form $*\varphi_0$ [Bry87].

DEFINITION 8.3 ([Joy00, §10.1, Def. 10.1.2]). A three-form φ on an oriented seven-manifold M is *positive* if at every $p \in M$ some oriented isomorphism $T_p M \rightarrow \mathbb{R}^7$ carries $\varphi|_p$ to φ_0 . Each positive φ determines a metric g_φ and four-form $*_\varphi \varphi$ via the pointwise identifications. We call the pair (φ, g_φ) a G_2 -structure. Such a pair is called *torsion-free* if $\nabla \varphi = 0$ for the Levi-Civita connection of g_φ . A G_2 -manifold is a triple (M, φ, g) with (φ, g) torsion-free.

REMARK 8.4. The set of such positive three-forms is open.

PROPOSITION 8.5 ([Joy00, Prop. 10.1.3, Props. 10.1.4, 10.1.5, Thm. 10.2.1, Prop. 10.2.2]). Let (φ, g) be a G_2 -structure on a seven-manifold M .

- (i) (φ, g) is torsion-free if and only if $\text{Hol}(g) \subseteq G_2$ with φ the induced three-form, if and only if $d\varphi = 0$ and $d*_\varphi \varphi = 0$. The first equation is linear while the second is not ($\varphi \mapsto *_\varphi \varphi$ takes the star in the metric induced by φ).
- (ii) The three-forms split orthogonally into irreducible G_2 -representations $\Lambda^3 T^* M = \Lambda_1^3 \oplus \Lambda_7^3 \oplus \Lambda_{27}^3$, with $\Lambda_1^3 = \langle \varphi \rangle$ and subscripts the fibre dimensions. Write π_1^3 for the orthogonal projection onto Λ_1^3 .
- (iii) If $\text{Hol}(g) \subseteq G_2$ then g is Ricci-flat.
- (iv) If (φ, g) is torsion-free then $\text{Hol}(g)$ is one of $\{1\}$, $SU(2)$, $SU(3)$, G_2 , for the standard embeddings acting on $\mathbb{R}^7 = \mathbb{R}^3 \oplus \mathbb{C}^2$ and $\mathbb{R} \oplus \mathbb{C}^3$. If moreover M is compact, $\text{Hol}(g) = G_2$ if and only if $\pi_1(M)$ is finite.

The reduction to $SU(2) \subset G_2$ arises from products. This is where the Eguchi-Hanson metric is used.

PROPOSITION 8.6 ([Joy00, Prop. 11.1.1]). Let (Y, g_Y) be hyperkähler, with hyperkähler triple $(\omega_1, \omega_2, \omega_3)$. Write $\omega = \omega_1$ and $\Theta = \omega_2 + i\omega_3$. On $\mathbb{R}^3 \times Y$ (or $T^3 \times Y$) with flat coordinates x_i set $g = (dx_1^2 + dx_2^2 + dx_3^2) \times g_Y$ and

$$\varphi = dx_{123} + dx_1 \wedge \omega + dx_2 \wedge \text{Re } \Theta - dx_3 \wedge \text{Im } \Theta. \quad (8.3)$$

Then (φ, g) is a torsion-free G_2 -structure, with $\text{Hol}(g) \subseteq \text{Sp}(1) \subset G_2$.

Sketch. At each point an orthonormal frame gives (ω, Θ) in standard constant form and (8.3) reproduces φ_0 , so φ is positive with metric g . Since dx_i, ω, Θ are closed so are φ and $*_\varphi \varphi$ (computed in [Joy00, eq. (11.2)]), and Proposition 8.5(i) applies. $\square$

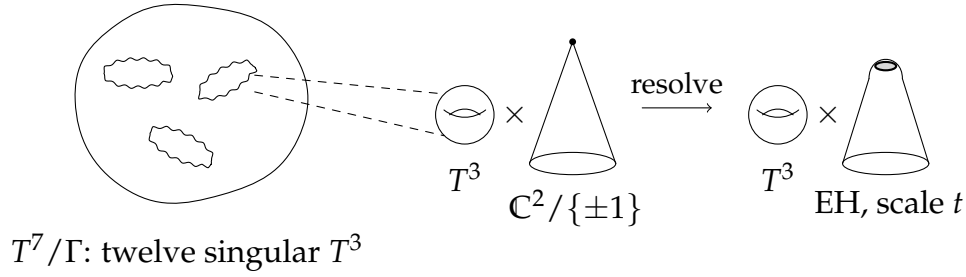


Figure 8.2: Joyce's construction. The singular set of T^7/Γ is twelve disjoint flat three-tori, each with a tubular neighbourhood isometric to $T^3 \times \mathbb{C}^2/\{\pm 1\}$. Each is replaced by the product of the flat T^3 with an Eguchi-Hanson space with central two-sphere at scale t , giving the closed G_2 -structures φ_t to which Theorem 8.7 applies.

The orbifold T^7/Γ and its resolution

Take $T^7 = \mathbb{R}^7/\mathbb{Z}^7$ with the flat structure (8.2), and let $\Gamma = \langle \alpha, \beta, \gamma \rangle \cong \mathbb{Z}_2^3$, each generator changing the signs of four coordinates in patterns preserving φ_0 (for instance α negates $x_4, \dots, x_7$), with half-period translations so that only α, β, γ themselves have fixed points [Joy00, §12.2]. Each generator fixes 16 three-tori, on which the group generated by the other two acts freely, leaving 4 each: the singular set is 12 disjoint copies of T^3 , each with neighbourhood $T^3 \times \mathbb{C}^2/\{\pm 1\}$ (Figure 8.2). The transverse singularity is again the ordinary double point, and the repair is the product of the flat T^3 with the Eguchi-Hanson space with central two-sphere at scale t , made into a torsion-free G_2 -structure by Proposition 8.6. Resolving all twelve tori produces a compact, simply connected seven-manifold M with $b^2 = 12$, $b^3 = 43$ [Joy00, §12.2], and interpolating with cut-offs gives closed G_2 -structures φ_t on M . The second equation fails on the gluing annuli to order $O(t^4)$.

The existence theorem

THEOREM 8.7 (Joyce [Joy96]; see [Joy00, Thm. 11.6.1]). *Let $\lambda, \mu, \nu > 0$. Then there exist $\kappa, K > 0$ such that the following holds for $0 < t \leq \kappa$. Let (φ, g) be a G_2 -structure on a compact seven-manifold M with $d\varphi = 0$, and suppose ψ is a smooth three-form with $d^*\varphi = d^*\psi$, satisfying $\|\psi\|_{L^2} \leq \lambda t^4$, $\|\psi\|_{C^0} \leq \lambda t^{1/2}$ and $\|d^*\psi\|_{L^{14}} \leq \lambda$, while the injectivity radius and curvature of g satisfy $\delta(g) \geq \mu t$ and $\|R(g)\|_{C^0} \leq \nu t^{-2}$. Then M carries a smooth torsion-free G_2 -structure $(\tilde{\varphi}, \tilde{g})$ with $\|\tilde{\varphi} - \varphi\|_{C^0} \leq K t^{1/2}$.*

The proof is the same deformation argument as in the Kummer construction, now applied to the three-form ψ rather than to a Kähler potential: the glued structure is an approximate solution, deformed within its cohomology class to an exact solution. One seeks $\tilde{\varphi} = \varphi + d\eta$ for η a two-form. The remaining condition

is $d(*_{\varphi+d\eta}(\varphi+d\eta)) = 0$, the Hodge star taken in the metric of the perturbed structure, and Joyce shows it is equivalent to

$$\Delta\eta = d^*\psi + d^*(f\psi) + *dF(d\eta), \quad f\varphi = \frac{7}{3}\pi_1^3(d\eta), \quad (8.4)$$

where $\Delta = dd^* + d^*d$ is the Hodge Laplacian, π_1^3 is the projection of Proposition 8.5(ii), and F vanishes to second order (this is [Joy00, eq. (11.33)]). Since $\text{Hol} = G_2$ if and only if $\pi_1(M)$ is finite (Proposition 8.5(iv)), and the M above are simply connected, these are the first compact manifolds with holonomy G_2 . One notes that both constructions meet only the singularity $\mathbb{C}^2/\{\pm 1\}$, transverse to the singular set, worse singularities require the other ALE spaces of Kronheimer's classification (Remark 6.9).

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